

02. Visualization Typology

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<http://info-design-lab.github.io>

Why visualize?

To convey information through visual representations in order to:

- Map/record/summarise
- Clarify/explain
- Instruct/teach/communicate
- Discover
- Inspire/advocate/persuade/tell a story

I. Classification based on the form

Temporal data visualizations track time series data – the performance of an indicator over a period of time. They normally feature lines that either stand alone or overlap with each other, with a start and finish time.

- Line graph, bar chart, gantt chart, area chart, scatter plots
- Polar area diagrams (Coxcomb chart, Florence Nightingale)
- Time series sequences
- Timelines
- Line graphs

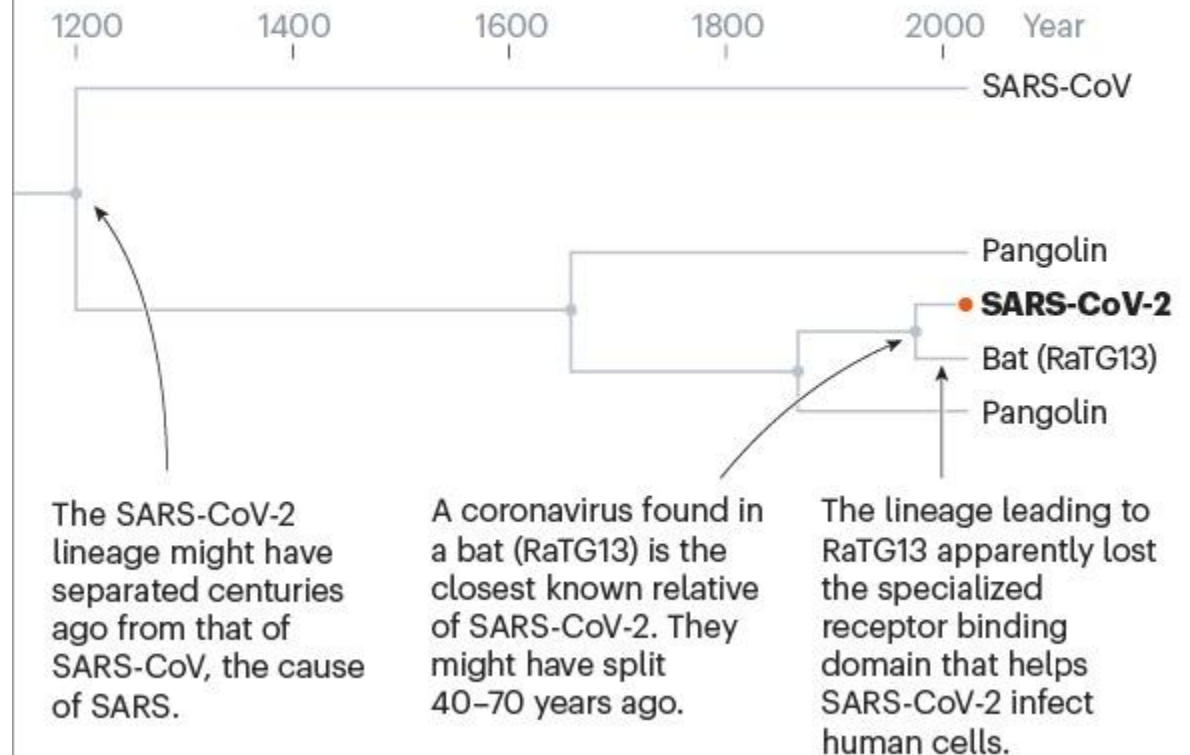
Classification based on the form

Hierarchical data visualizations order groups within larger groups. They are best suited if you're looking to display clusters of information, especially if they flow from a single origin point.

- Tree diagrams
(<https://info-design-lab.github.io/UDISE/>)
- Ring charts
- Sunburst diagrams

A FAMILY OF KILLERS

Researchers are trying to work out the evolutionary relationships between the new coronavirus (SARS-CoV-2) and others in a large family of related viruses (a simplified version is shown below). A study, which has not yet been peer reviewed, suggests a possible scenario for how SARS-CoV-2 evolved.



Classification based on the form

Network Datasets connect deeply with other datasets. Network data visualizations show how they relate to one another within a network. In other words, demonstrating relationships between datasets without wordy explanations.

- Matrix charts
- Node-link diagrams
- Word clouds
- Alluvial diagrams (Sankey diagram - <https://info-design-lab.github.io/accident/>)

You can find more form based classes at
https://guides.library.duke.edu/datavis/vis_types

Classification based on the form

Geospatial relate to real life physical locations, overlaying familiar maps with different data points.

- Flow map (Minard's Napoleon's March)
- Density map
- Cartogram
- Heat map

II. Classification based on intention (analytical, presentation)

A visualization is recommended to users by satisfying their requirements for a particular dataset. There have been many approaches like rule-based, machine learning approaches, and knowledge graph embedding approaches for recommending the visualization for the dataset. These approaches recommend visualization based on the rules, training a machine learning model by extracting features of data and by constructing a knowledge graph from the data respectively.

We consider a conversational approach to visualization, where users specify their needs at each step in natural language, with a visualization being returned in turn. Prior work has shown that visualization generation can be boiled down to the identification of visualization intent and visual encodings.

II. Classification based on intention (analytical, presentation)

Trends time series (using line or bar charts), motion chart, scatter plot

Correlation scatter plot, bubble chart, heat map

Distribution histogram, boxplot, ridgelines, violin chart, density map

Part to whole pie chart, donut chart, stacked bar chart, funnel chart

Ranking bar plot, spider/radar chart, word cloud, parallel coordinates, lollipop chart

Spatial relationship (Geo) choropleth, cartogram, hexbin map, route map

Flow chord diagram, sankey diagram, arc diagram, organisation chart



from Data to Viz

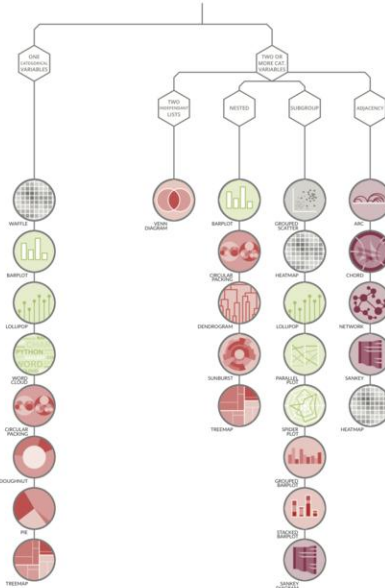
'From Data to Viz' is a classification of chart types based on input data format. It will help you find the perfect chart in three simple steps:

- 1 Identify what type of data you have.
- 2 Go to the corresponding decision tree and follow it down to a set of possible charts.
- 3 Choose the chart from the set that will suit your data and your needs best.

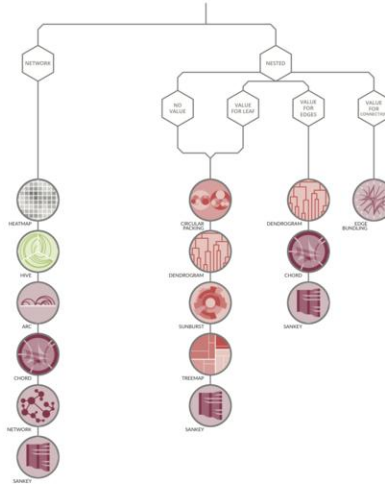
Dataviz is a world with endless possibilities and this project does not claim to be exhaustive. However it should provide you with a good starting point. For an interactive version and much more, visit:

data-to-viz.com

CATEGORIC



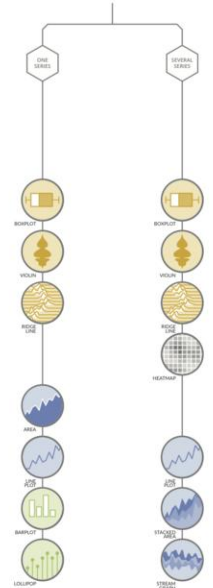
RELATIONAL



MAP



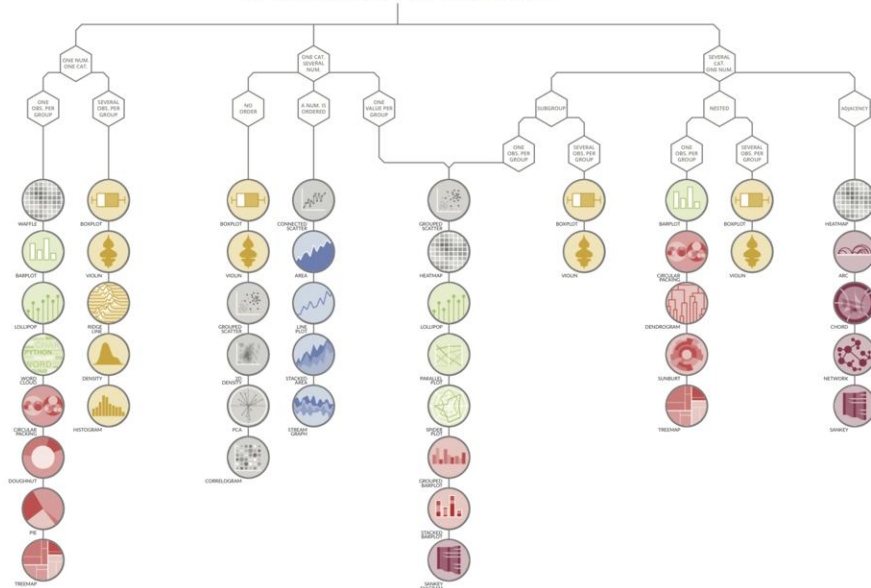
TIME SERIES



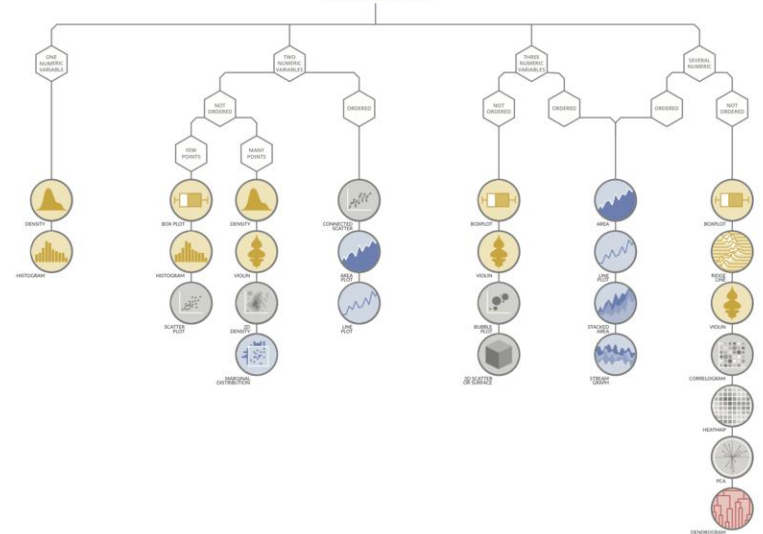
WHAT DO YOU WANT TO SHOW ?

- Distribution
- Correlation
- Ranking
- Part of a whole
- Evolution
- Maps
- Flow

CATEGORIC AND NUMERIC



NUMERIC



III. Classification based on user experience

Narratives The objective is to explain by giving the reader a vicarious experience of the intent through a story. Narrative information graphics are characterized by a story told with a distinct point of view.

Instructives explain step by step how things work. Here, the objective is to explain, by enabling the reader to sequentially step through the intent.

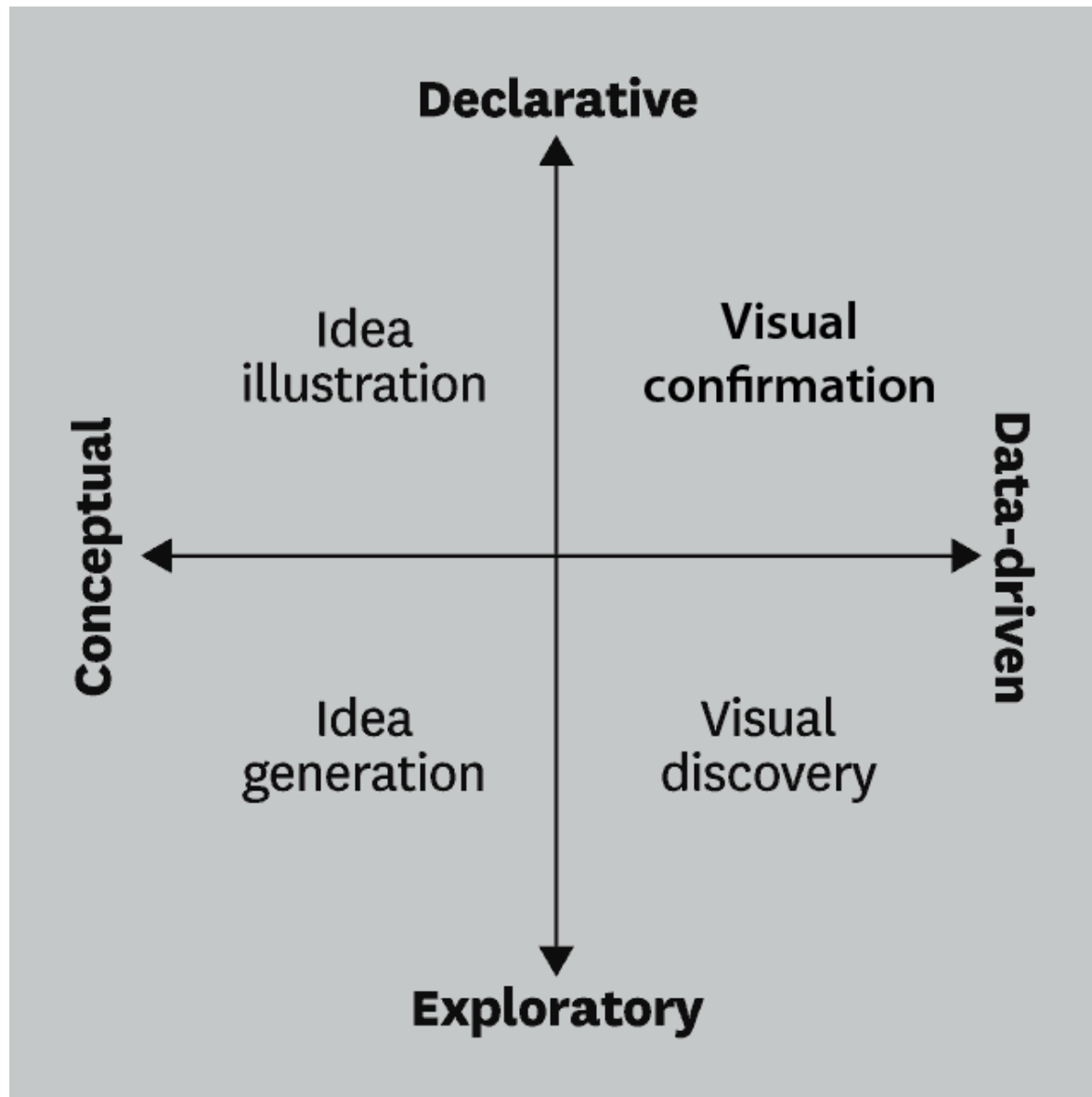
Exploratives offer the reader an opportunity to explore and discover the intent.

Simulatives allow the reader 'to experience the intent (usually a real world phenomena).

(Nichani and Rajamanickam 2003).

III. The 4 types of Data Visualization:

Scott Berinato, HBR, June 2016



1. Idea Illustration

Idea illustrations clarify complex ideas by drawing on our ability to understand directions (up, down, forward, backward), metaphors (trees, bridges) and simple design conventions (circles, hierarchies). Org charts and decision trees are classic examples of idea illustration.

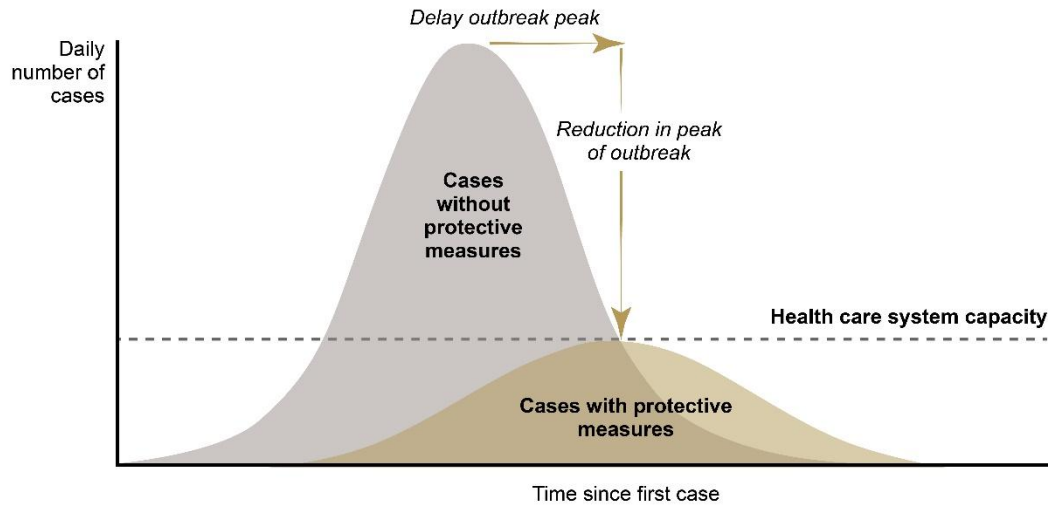
INFO TYPE: Process, framework

TYPICAL SETTING: Presentations, teaching

PRIMARY SKILLS: Design, editing

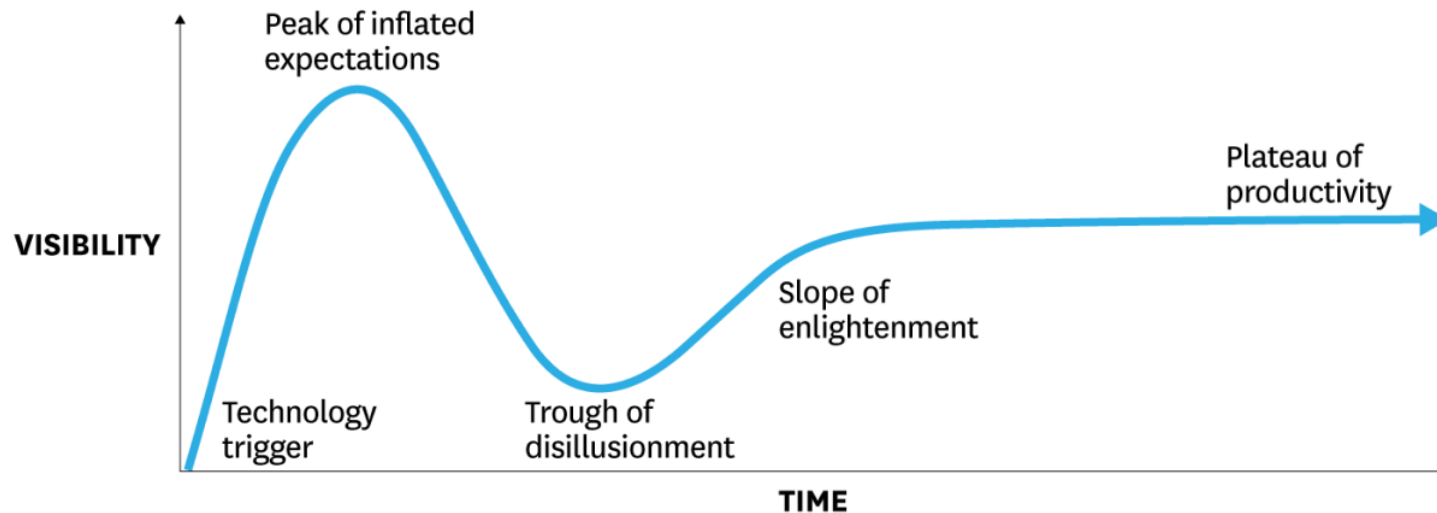
GOALS: Learning, simplifying, explaining

FLATTENING THE CURVE



Source: CDC

Hype Cycle for Emerging Technologies



Fats, Oils, & Sweets
Use Sparingly



Milk, Yogurt & Cheese Group
2-3 Servings



Meat, Poultry, Fish, Dry
Beans, Eggs, & Nuts Group
2-3 Servings



Vegetable Group
3-5 Servings



Fruit Group
2-4 Servings



Bread, Cereal,
Rice, & Pasta
Group
6-11
Servings



2. Idea Generation

Visualization as a tool to support idea generation. Like idea illustration, idea generation relies on conceptual metaphors, but it takes place in more-informal settings, such as off-sites, strategy sessions, and early-phase innovation projects.

INFO TYPE: Complex, undefined

TYPICAL SETTING: Working session, brainstorming

PRIMARY SKILLS: Team-building, facilitation

GOALS: Problem solving, discovery, innovation

①

LOTS OF CUSTOMERS
SPEND A LITTLE



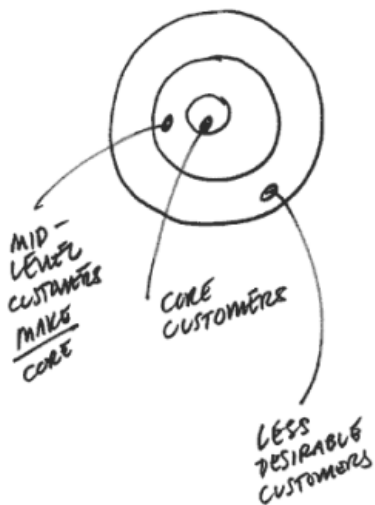
FEWER CUSTOMERS
SPENDING MORE!



20 @ \$120 PER CUSTOMER
= \$2,400



6 @ \$500
= \$3,000



②

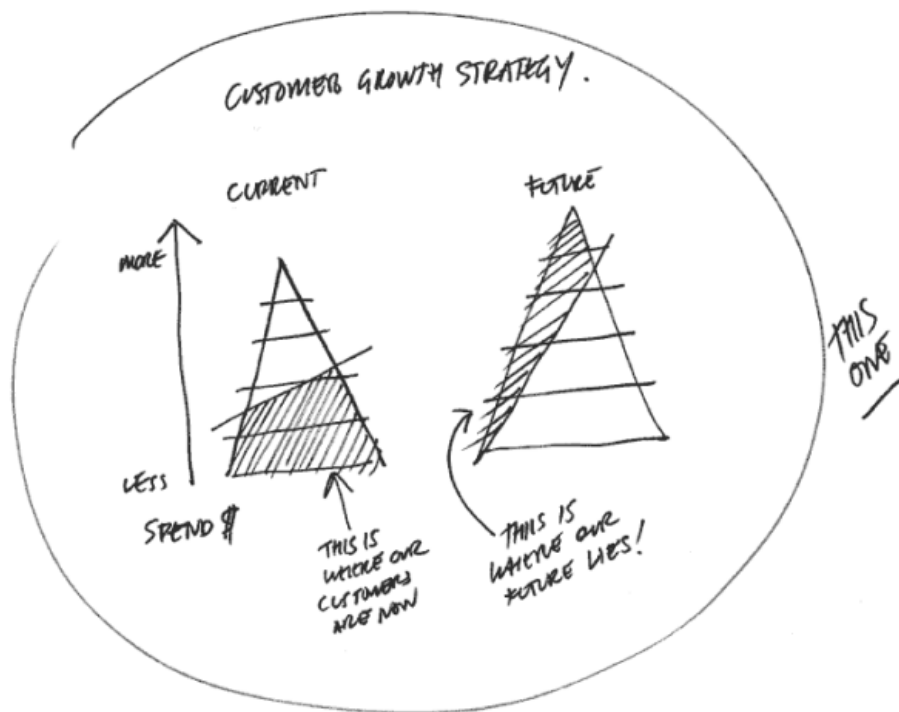
US . NOW

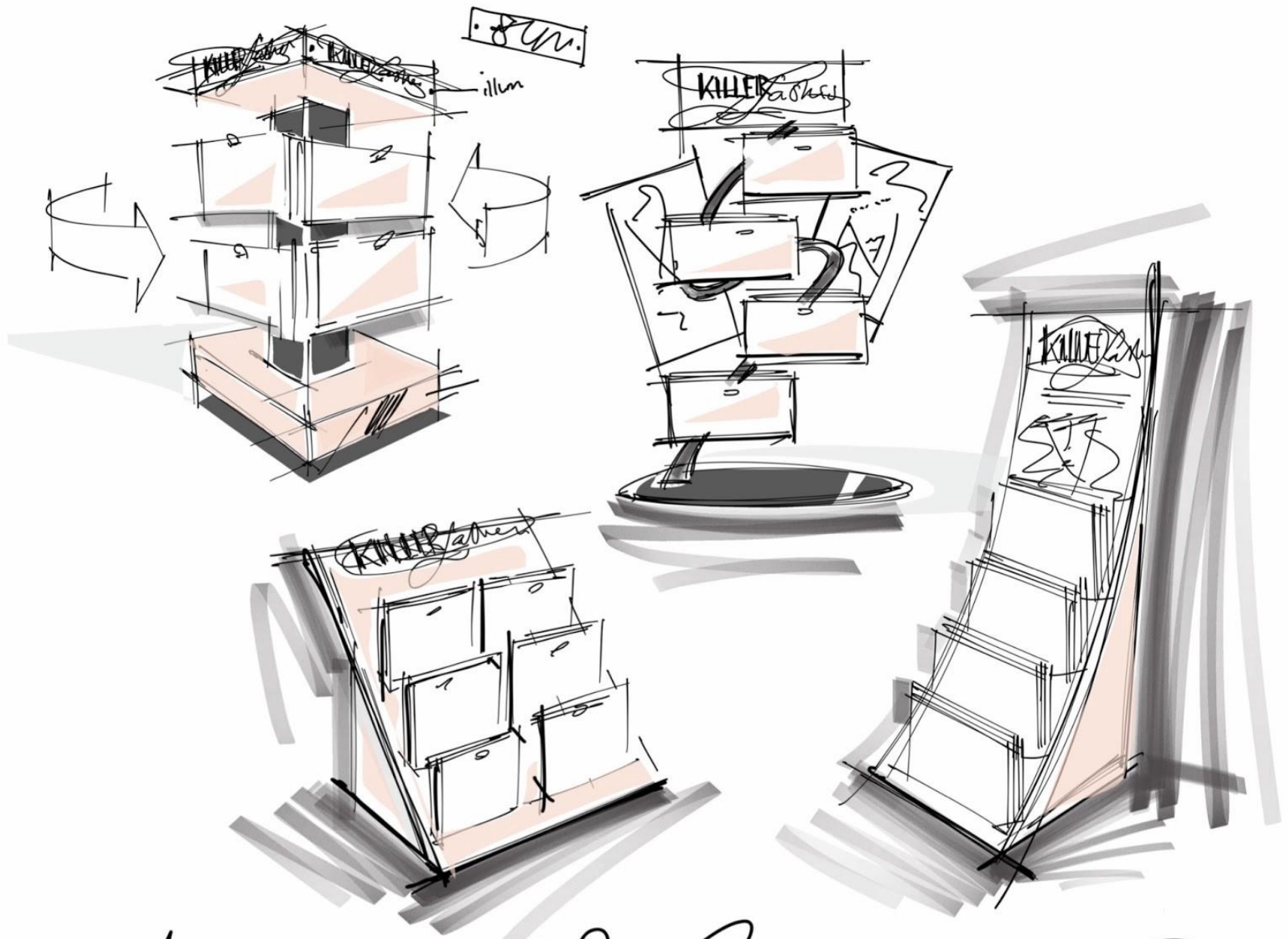


FUTURE?



BIGGER
PIE.
FEWER
SLICES.?





KILLER Lattes Counter Units.

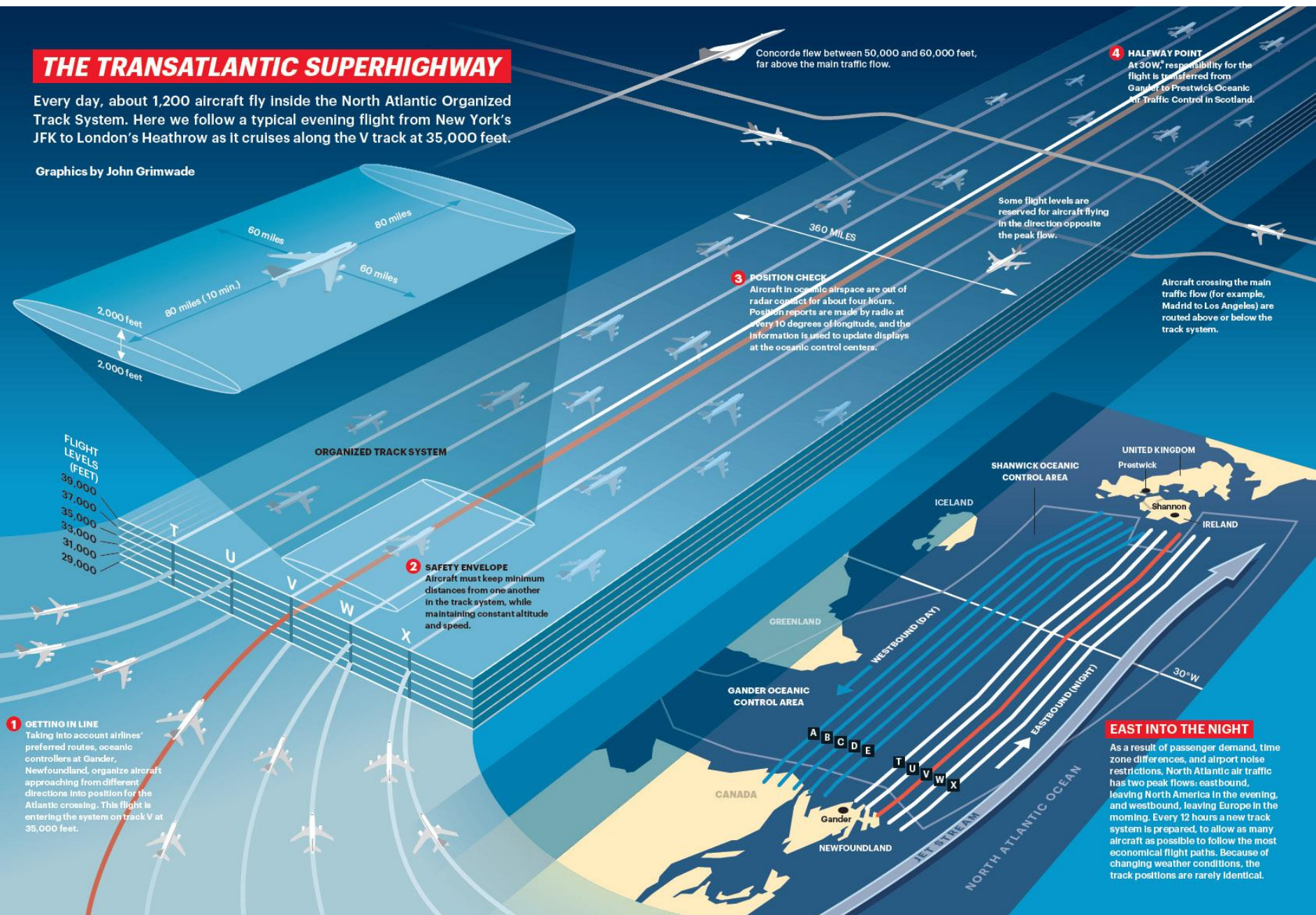
3. Visual Confirmation

You're answering one of two questions with this kind of project: Is what I suspect actually true? or What are some other ways of depicting this idea?

THE TRANSATLANTIC SUPERHIGHWAY

Every day, about 1,200 aircraft fly inside the North Atlantic Organized Track System. Here we follow a typical evening flight from New York's JFK to London's Heathrow as it cruises along the V track at 35,000 feet.

Graphics by John Grimwade



Concorde flew between 50,000 and 60,000 feet, far above the main traffic flow.

4 HALFWAY POINT
At 30°W, responsibility for the flight is transferred from Gander to Prestwick Oceanic ATC in Scotland.

Some flight levels are reserved for aircraft flying in the direction opposite the peak flow.

3 POSITION CHECK
Aircraft in oceanic airspace are out of radar contact for about four hours. Position reports are made by radio at every 10 degrees of longitude, and the information is used to update displays at the oceanic control centers.

Aircraft crossing the main traffic flow (for example, Madrid to Los Angeles) are routed above or below the track system.

FLIGHT LEVELS (FEET)
39,000
37,000
35,000
33,000
31,000
29,000

ORGANIZED TRACK SYSTEM

2 SAFETY ENVELOPE
Aircraft must keep minimum distances from one another in the track system, while maintaining constant altitude and speed.

1 GETTING IN LINE
Taking into account airlines' preferred routes, oceanic controllers at Gander, Newfoundland, organize aircraft approaching from different directions into position for the Atlantic crossing. This flight is entering the system on track V at 35,000 feet.

UNITED KINGDOM
Prestwick
Shannon
IRELAND
ICELAND
GREENLAND
GANDER OCEANIC CONTROL AREA
CANADA
NEWFOUNDLAND
Gander
JET STREAM
NORTH ATLANTIC OCEAN
30°W
WESTBOUND (DAY)
EASTBOUND (NIGHT)
A B C D E
T U V W X

EAST INTO THE NIGHT

As a result of passenger demand, time zone differences, and airport noise restrictions, North Atlantic air traffic has two peak flows: eastbound, leaving North America in the evening, and westbound, leaving Europe in the morning. Every 12 hours a new track system is prepared, to allow as many aircraft as possible to follow the most economical flight paths. Because of changing weather conditions, the track positions are rarely identical.

SUPERintrigante

Perguntas instigantes, respostas surpreendentes

BOCA EM AÇÃO COMO É O TRÁFICO NA FAVELA?

Os pontos de tráfico de drogas, conhecidos como "bocas", operam como empresas, escondidos em favelas e bairros pobres das grandes cidades. Os criminosos se organizam em uma hierarquia preocupada em garantir duas coisas: o abastecimento constante de cocaína, maconha e outros entorpecentes e o sistema de proteção contra a polícia ou quadrilhas rivais.

Para garantir a eficiência do negócio, são contratados diversos funcionários. O esquema de segurança e a acirrada disputa entre traficantes põem em risco a vida de compradores e moradores da favela. "Até chegar à boca, o usuário tem que andar na favela. Ele é avaliado e nem percebe. Se os seguranças pensarem que ele é um policial disfarçado, atiram", diz o delegado Carlos Roberto Alves de Andrade, da Delegacia de Repressão ao Crime Organizado do Departamento de Narcóticos de São Paulo. **5**

por: Cássio da Silva

CRÍME ORGANIZADO VÁRIOS FUNCIONÁRIOS ESTÃO ENVOLVIDOS NO ESQUEMA DE TRÁFICO

AVIOZINHOS

Os garotos que levam a droga da boca para os clientes são mais comuns no Rio de Janeiro. Em São Paulo, onde as favelas são planas, a distância entre a boca e o consumidor é pequena e o serviço deles nem sempre é necessário.

ALTO ESCALÃO

Traficantes de maior hierarquia ficam posicionados sobre lajes e barracos, onde podem se proteger melhor e atirar em caso de tentativa de invasão. Carregam fuzis, ideais para combates a longa distância.

A BOCA

Geralmente fica perto de riachos, esgotos ou barrancos, para dificultar a chegada da polícia. Em uma mesma favela, podem existir várias bocas e nem toda a droga fica aqui. Barracos conhecidos como "paiol" são usados para armazenamento de grandes quantidades e da munição da quadrilha.

GERENTE DA BOCA

É responsável pela chegada da droga e pela contratação do pessoal. É ele que comanda toda a operação dentro da favela e, por isso, é sempre alguém de muita confiança do dono da boca.

SEGURANÇAS

A função deles é proteger os amadores da boca da polícia e de traficantes rivais. Eles usam armas próprias para combater a curta distância.

ENQUANTO ISSO...

O dono da "boca" não lida diretamente com a venda da droga. Ele comanda o tráfico de um barraco ou casa afastada, por meio dos gerentes. Bocas bem-sucedidas podem transformar traficantes em homens ricos e bem de vida.

4. Visual Exploration

Open-ended data-driven visualizations It is intended to produces insights that can't be gleaned any other way.

INFO TYPE: Big data, complex, dynamic

TYPICAL SETTING: Working sessions, testing, analysis

PRIMARY SKILLS: Business intelligence, programming, paired analysis

GOALS: Trend spotting, sense making, deep analysis

A PERIODIC TABLE OF VISUALIZATION METHODS

C continuum	Data Visualization Visual representations of quantitative data in schematic form (either with or without axes)										G graphic facilitation						
Tb table	Ca cartesian coordinates	Information Visualization The use of interactive visual representations of data to amplify cognition. This means that the data is transformed into an image, it is mapped to screen space. The image can be changed by users as they proceed working with it										Mm metro map	Tr tree	Ct cartoon			
Pi pie chart	L line chart	Concept Visualization Methods to elaborate (mostly) qualitative concepts, ideas, plans, and analyses.										Co communication diagram	Fp flight plan	Cs concept skeleton	Br bridge	Fu funnel	Ri rich picture
B bar chart	Ac area chart	R radar chart	Pa parallel coordinates	Hy hyperbolic tree	Cy cycle diagram	T timeline	Ve venn diagram	Mi mindmap	Sq square of oppositions	Cc concentric circles	Ar argument slide	Sw swim lane diagram	Gc gantt chart	Pm perspectives diagram	D dilemma diagram	Pr parameter ruler	Kn knowledge map
Hi histogram	Sc scatterplot	Sa sankey diagram	In information lense	E entity relationship diagram	Pt petri net	Fl flow chart	Cl clustering	Lc layer chart	Py minto pyramid technique	Ce cause-effect chains	Tl toulmin map	Dt decision tree	Cp cpm critical path method	Cf concept fan	Co concept map	Ic iceberg	Lm learning map
Tk tukey box plot	Sp spectrogram	Da data map	Tp treemap	Cn cone tree	Sy system dyn./simulation	Df data flow diagram	Se semantic network	So soft system modeling	Sn synergy map	Fo force field diagram	Ib ibis argumentation map	Pr process event chains	Pe pert chart	Ev evocative knowledge map	V Vee diagram	Hh heaven 'n' hell chart	I informul

Cy

Process Visualization

Note: Depending on your location and connection speed it can take some time to load a pop-up picture.

version 1.5

© Ralph Lengler & Martin J. Eppler, www.visual-literacy.org

Hy

Structure Visualization

 $\angle$

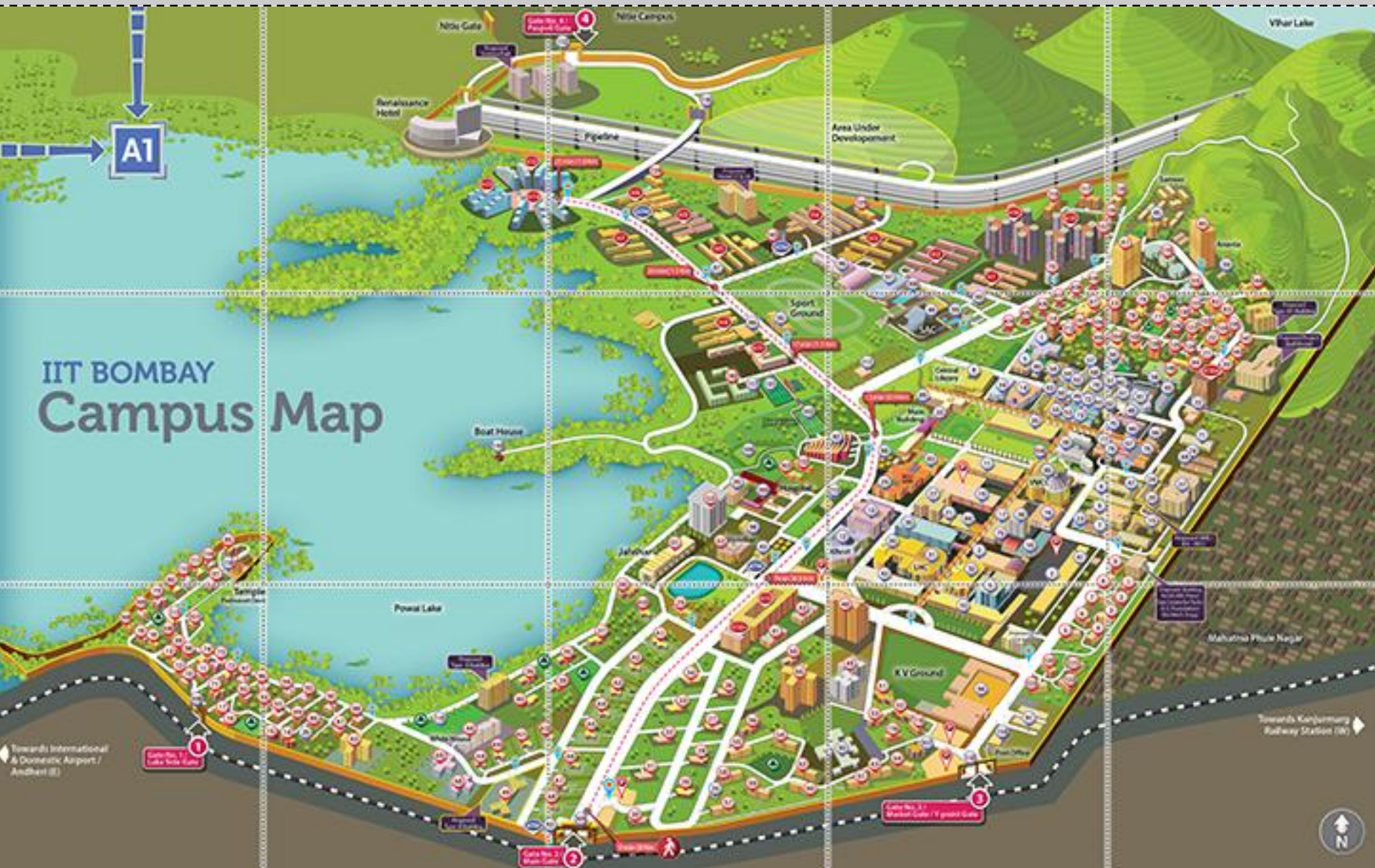
Overview
Detail

Detail AND Overview

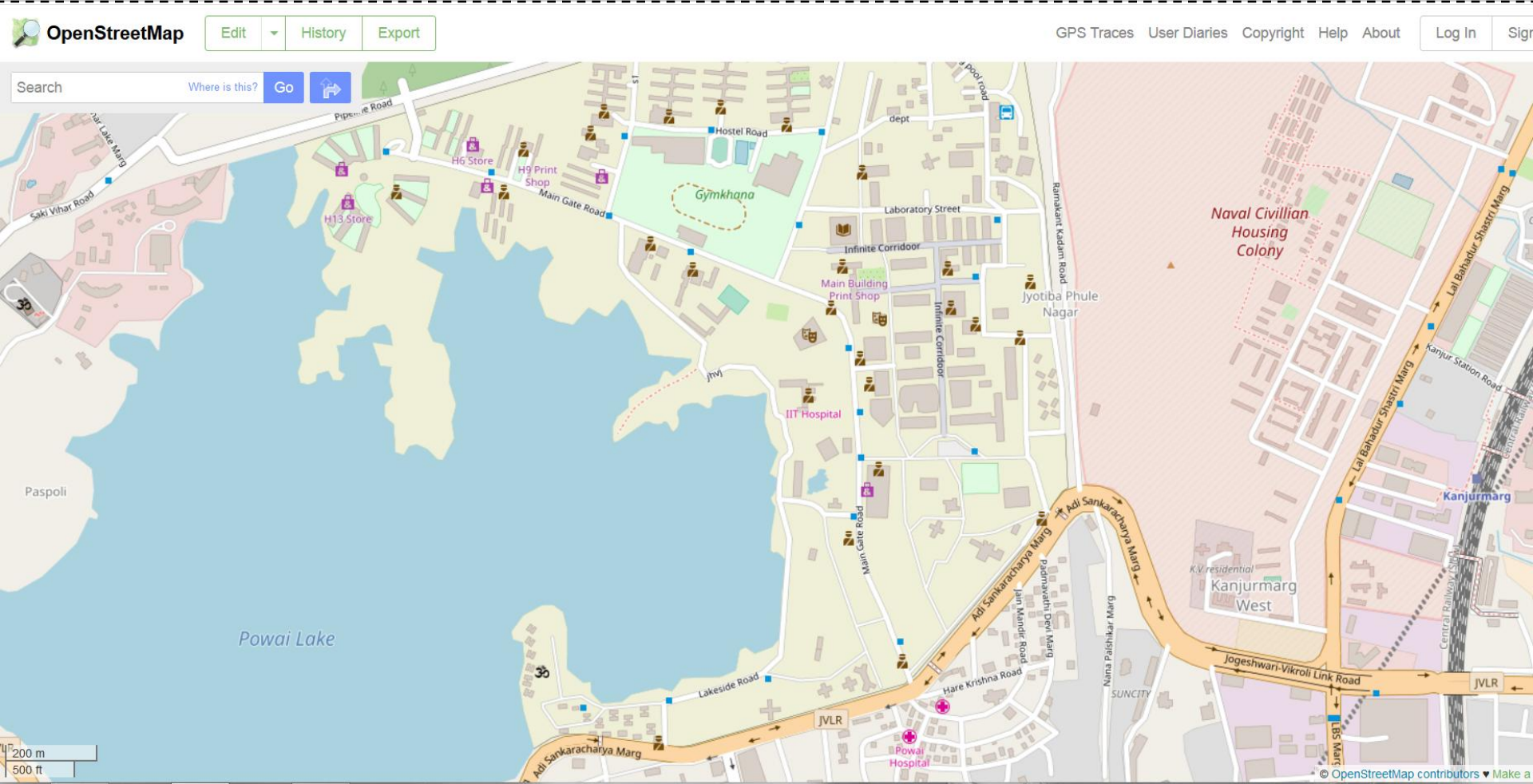
Divergent thinking

 Su supply demand curve	 Pc performance charting	 St strategy map	 Oc organisation chart	 Ho house of quality	 Fd feedback diagram	 Ft failure tree	 Mq magic quadrant	 Ld life-cycle diagram	 Po porter's five forces	 S s-cycle	 Sm stakeholder map	 Is ishikawa diagram	 Tc technology roadmap
 Ed edgeworth	 Pf portfolio	 Sg strategic	 Mz mintzberg's	 Z zwick's	 Ad affinity	 De decision	 Bm bcs matrix	 Stc strategy	 Vc value chain	 Hy hype-cycle	 Sr stakeholder	 Ta taps	 Sd sorav

Infographic Vs. Data Visualization



Infographic Vs. Data Visualization



Infographic Vs. Data Visualization

In their book *Designing Data Visualizations*, authors Noah Iliinsky and Julie Steele use the following three criteria to determine whether to call a graphic a data visualization or an infographic:

1. **Method of generation:** This criterion refers to what goes into creating the graphic itself. If lots of original illustrations are created to explain the data, for example, it's likely to be an infographic. You often see infographics with beautiful, elaborate images created to explain the information.

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2. **Quantity of data represented:** Typically, data visualizations have more and different kinds of data from infographics. Also, the data in data visualizations changes frequently to indicate changes in status. In addition, an infographic is less likely to include interactive numbers.

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2. **Quantity of data represented:** Typically, data visualizations have more and different kinds of data from infographics. Also, the data in data visualizations changes frequently to indicate changes in status. In addition, an infographic is less likely to include interactive numbers.
3. **Degree of aesthetic treatment applied:** This criterion refers to the artfulness of the graphic. If a lot of design work has gone into displaying information, the graphic is likely to be an infographic.

Infographics

Visual representation of data that is:

- manually drawn (and therefore a custom treatment of the information);
- specific to the data at hand (and therefore nontrivial to recreate with different data);
- aesthetically rich (strong visual content meant to draw the eye and hold interest);
- relatively data-poor (because each piece of information must be manually encoded).

Data Visualization

Visual representation of data that is:

- algorithmically drawn (may have custom touches but is largely rendered with the help of computerized methods);
- easy to regenerate with different data (the same form may be repurposed to represent different datasets with similar dimensions or characteristics);
- often aesthetically barren (data is not decorated);
- relatively data-rich (large volumes of data are welcome and viable, in contrast to infographics).

HEIGHT of MAN and BASE



PARTICIPANTS



TICKET PRICE

THEME



MOON PHASE (burn night)



EVENT/FESTIVAL ATTENDANCE



THEMECAMPS (placed)

FIRE CONCLAVE GROUPS

ART PROJECTS (registered)

BLM CITATIONS

RECYCLED ALUMINUM CANS

COSTS



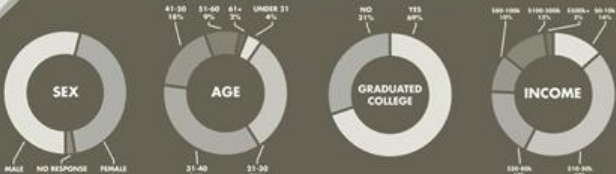
COST - PORTOPHONES



the TEMPLES of BURNING MAN



PARTICIPANT DEMOGRAPHICS



CITY POPULATIONS of NEVADA



AVERAGE WEEKLY WEATHER FORECAST



LOCAL POPULATION COMPARISON



BURNING MAN

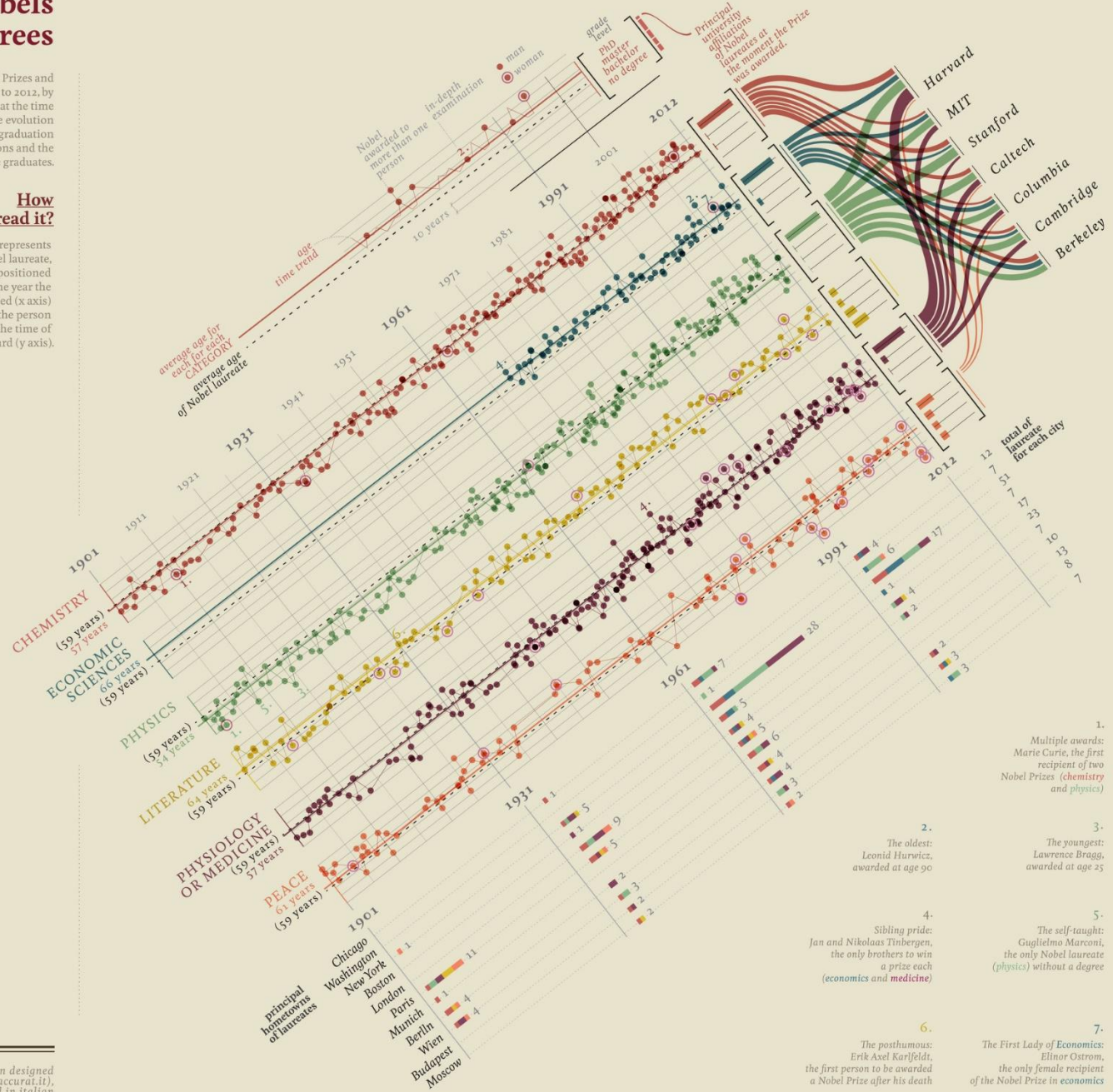
Black Rock Desert valley fills with water during Pleistocene ice age	Humboldt River flows to Black Rock Desert sub-basin	Lahontan lake covers 8,665 square miles; subsided under about 500 feet of water	Lake disappears significantly	Great Basin tribes inhabit area	Frémont Expedition	Peter Lassen led California Trail emigrants through Applegate-Lassen Corral	Western Pacific's Feather River Route starts (Oakland - Salt Lake City)	Largest mammoth found in North America	First Black Rock City built (Thruway)	Land speed record (Thruway)	First successful launch of an amateur rocket to space (Thruway)
70 MILLION YEARS AGO	15,000 YEARS AGO	12,700 YEARS AGO	9,000 YEARS AGO	7,000 YEARS AGO	1843	1846	1910	1978	1990	1997	2004

Nobels no degrees

This visualization explores Nobel Prizes and graduate qualifications from 1901 to 2012, by analysing the age of recipients at the time prizes were awarded, average age evolution through time and among categories, graduation grades, main university affiliations and the principal hometowns of the graduates.

How to read it?

Each dot represents a Nobel laureate, each recipient is positioned according to the year the prize was awarded (x axis) and age of the person at the time of the award (y axis).

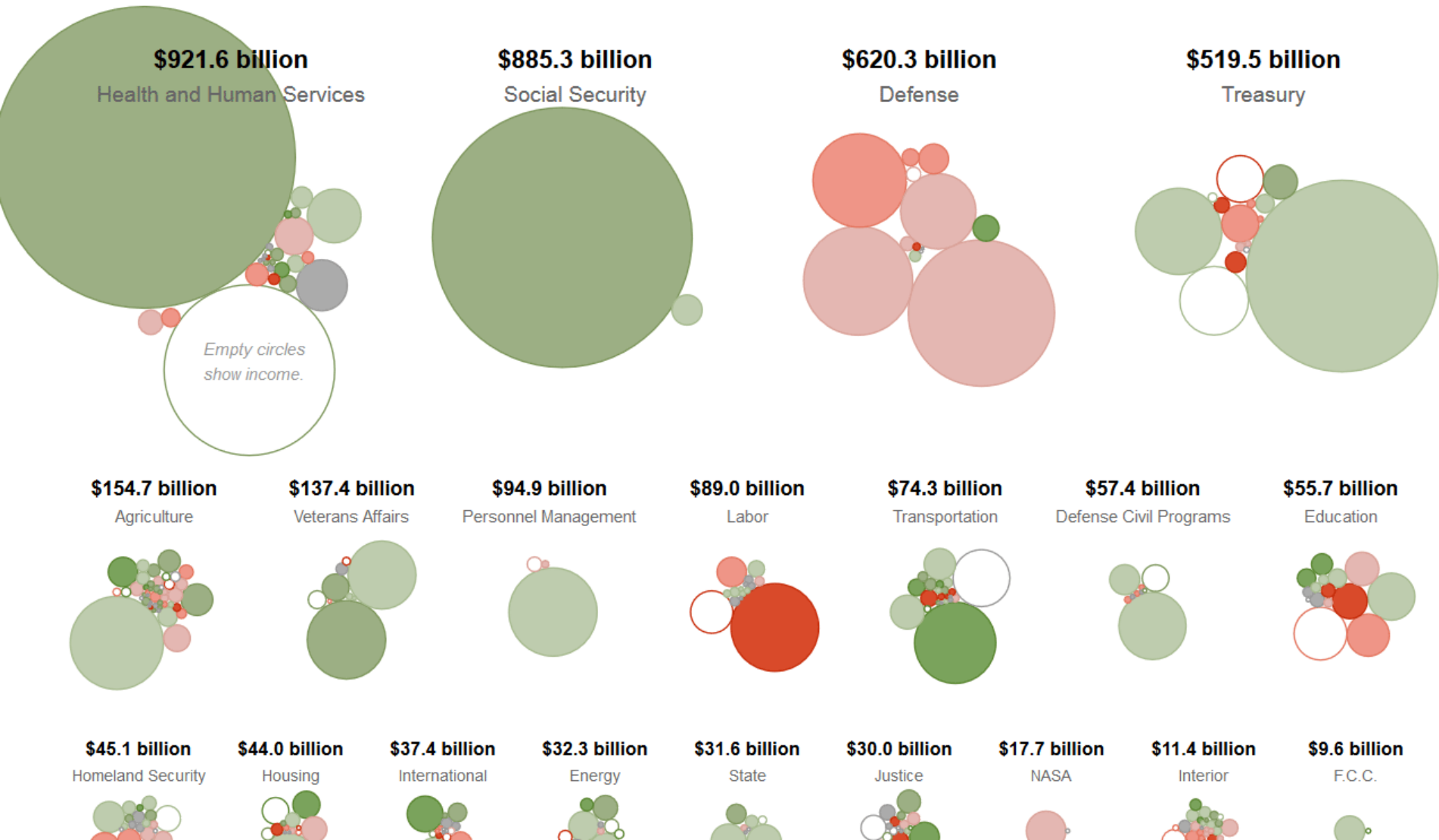


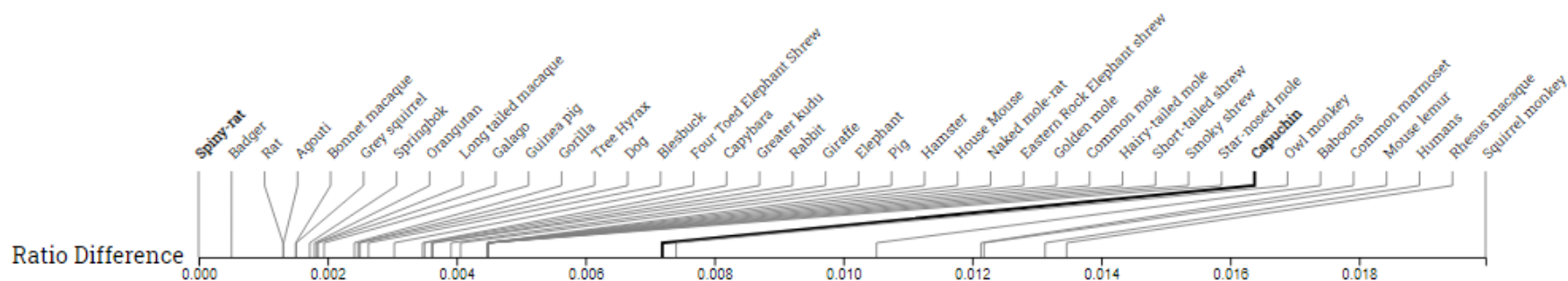
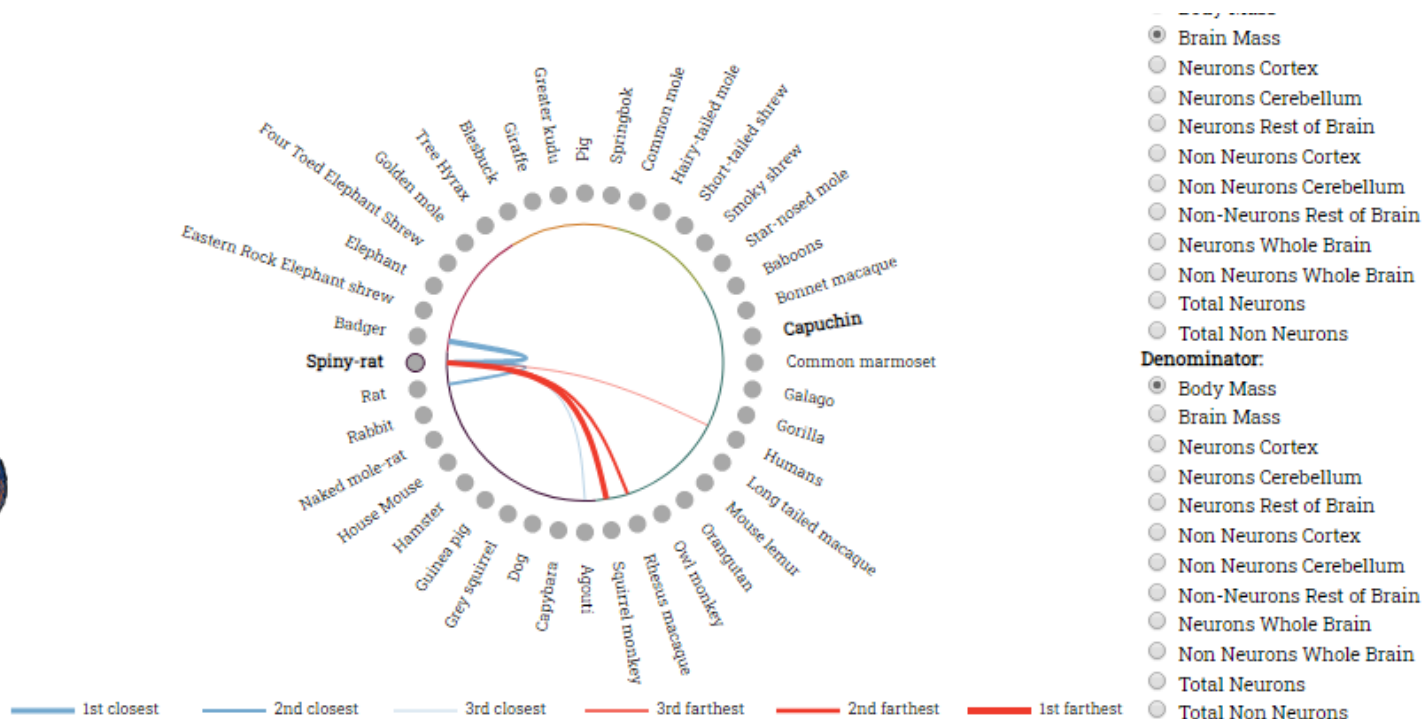
The visualization has been designed and produced by Accurat (www.accurat.it), and was originally published in Italian on La Lettura the Sunday cultural supplement of Corriere della Sera.

Four Ways to Slice Obama's 2013 Budget Proposal

Explore every nook and cranny of President Obama's federal budget proposal.

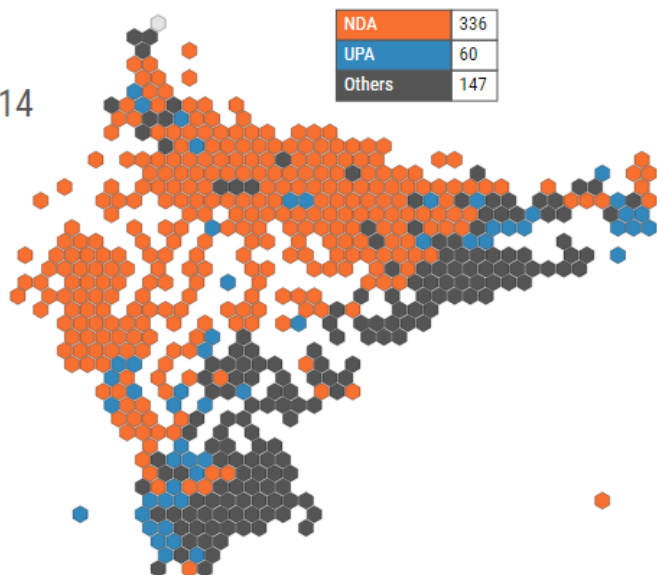
All Spending	Types of Spending	Changes	Department Totals
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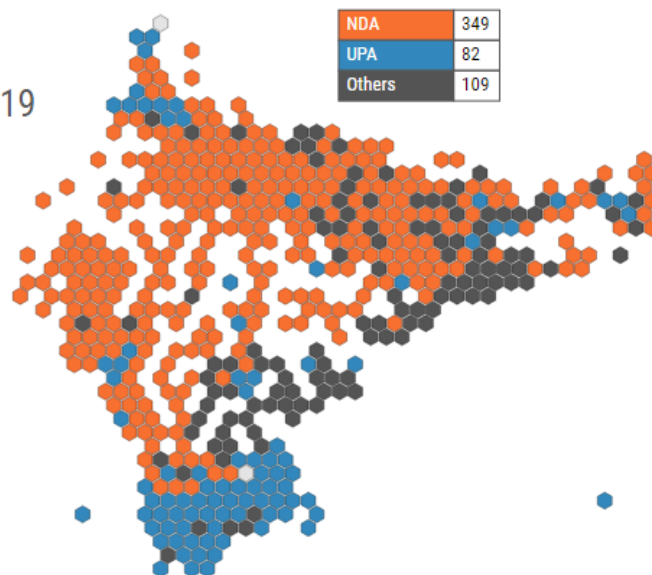


Cartogram ☒ Geographic Map

2014



2019



Alliances

State Map

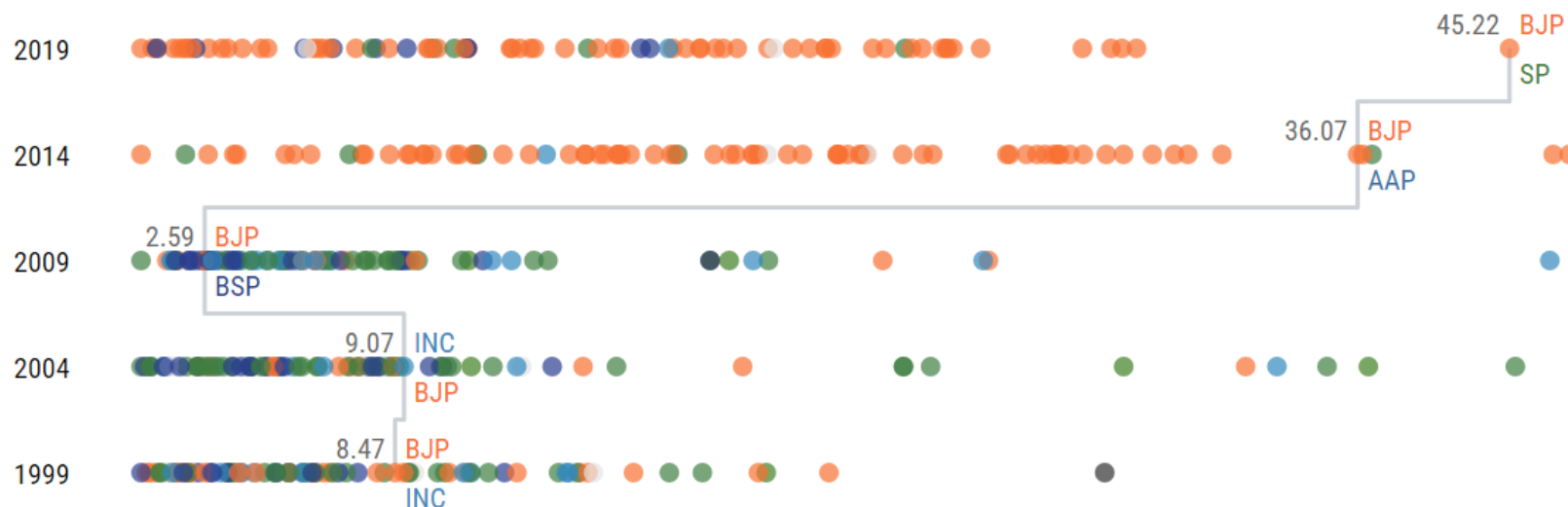
Voter Margin

General Election Results Explorer

State: UTTAR PRADESH

Constituency: VARANASI

Votes ☒ Percentage



Public Data

▼ World Development Indicators

▶ Economic Policy and Debt

▶ Education

▼ Environment

Agricultural irrigated land (% of tot...

Agricultural land (sq. km)

Agricultural machinery, tractors

Agricultural machinery, tractors pe...

Agricultural production index

Agriculture value added per worke...

Clear

▼ Compare by Country

Region Color

Lending Type All

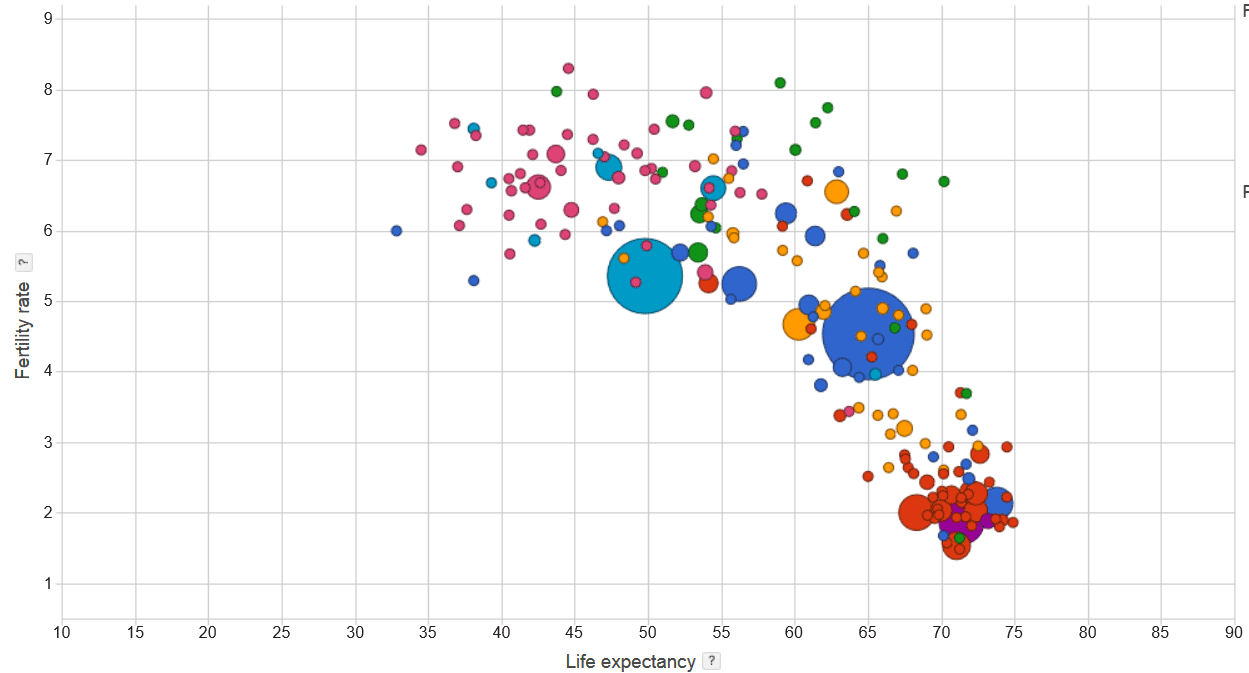
Income Level All

- ☐ Afghanistan
- ☐ Albania
- ☐ Algeria
- ☐ American Samoa
- ☐ Andorra
- ☐ Angola
- ☐ Antigua and Barbuda
- ☐ Argentina
- ☐ Armenia
- ☐ Aruba

Clear selections

- ▶ Gender
- ▶ Urbans/Rurals

Countries ?



1973

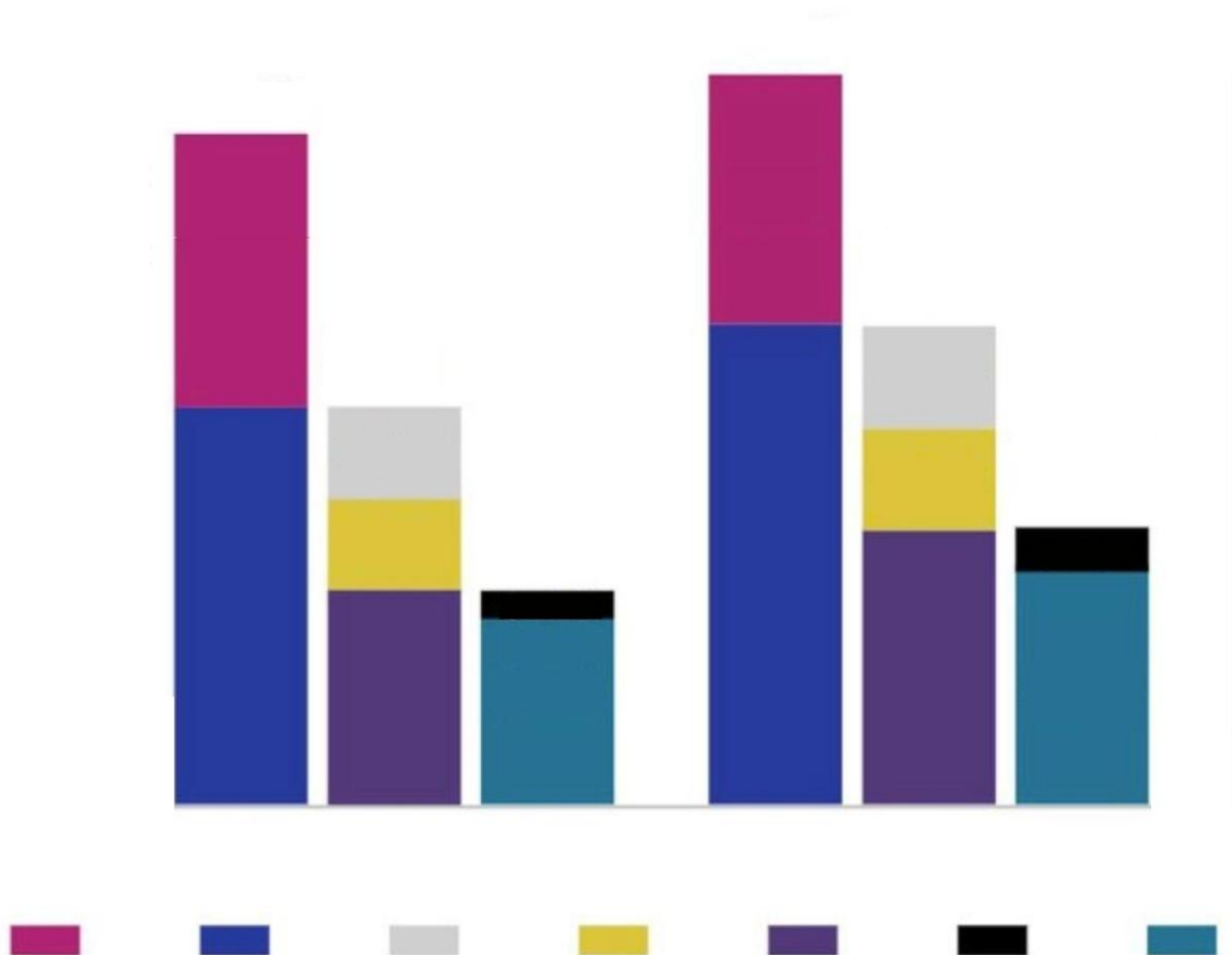
Data from World Bank Last updated: Jan 12, 2016

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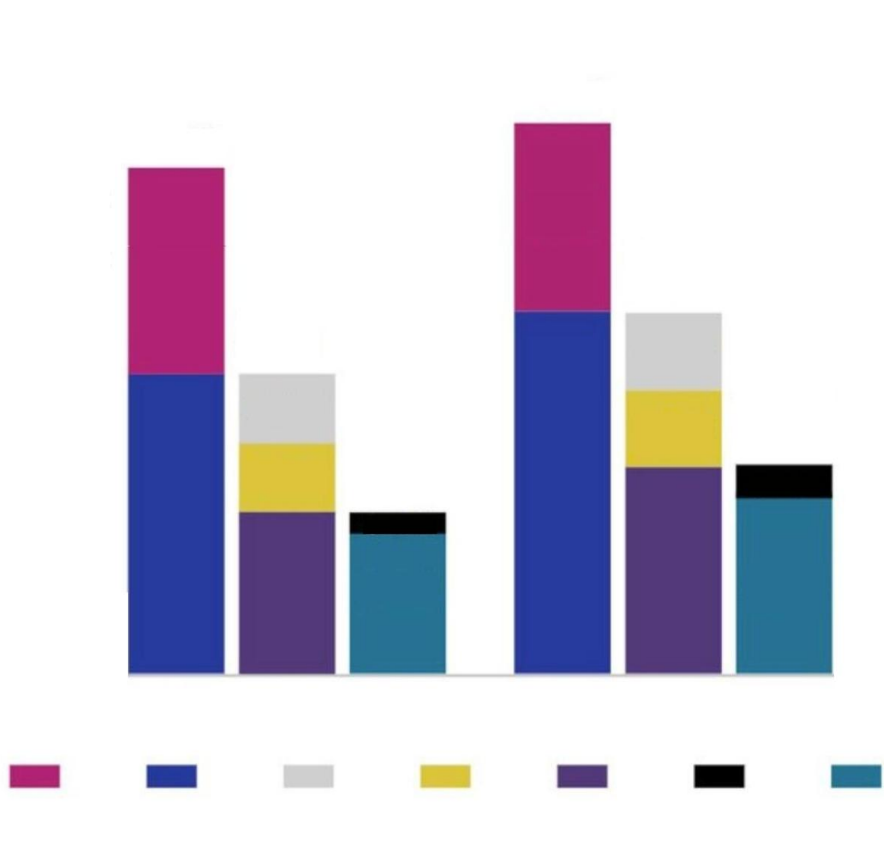
Assignment 2: Chart Deconstruction & Redesign

Deconstructing a chart amounts to a deeper version of the self-sufficiency test. It helps us understand what parts of the chart are doing what, and what cues certain parts send to readers.

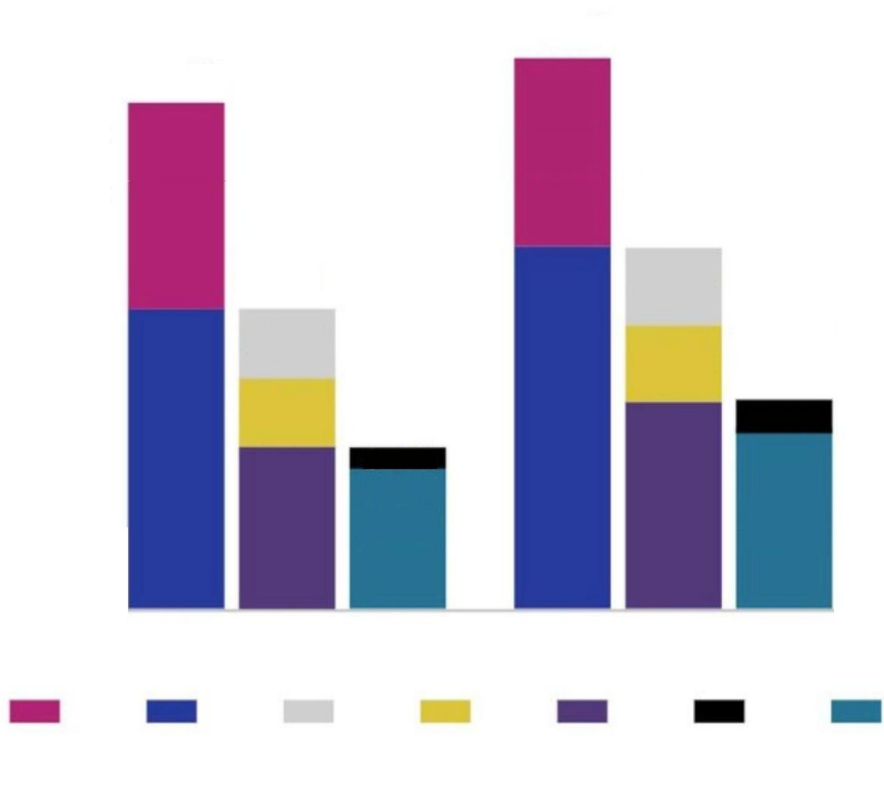
In better dataviz, every part of the chart is doing something useful, it's free of redundant parts that take up cognitive processing time for no reason, and the cues to readers move them towards the intended message, not away from it.



What can you guess? **Structure**

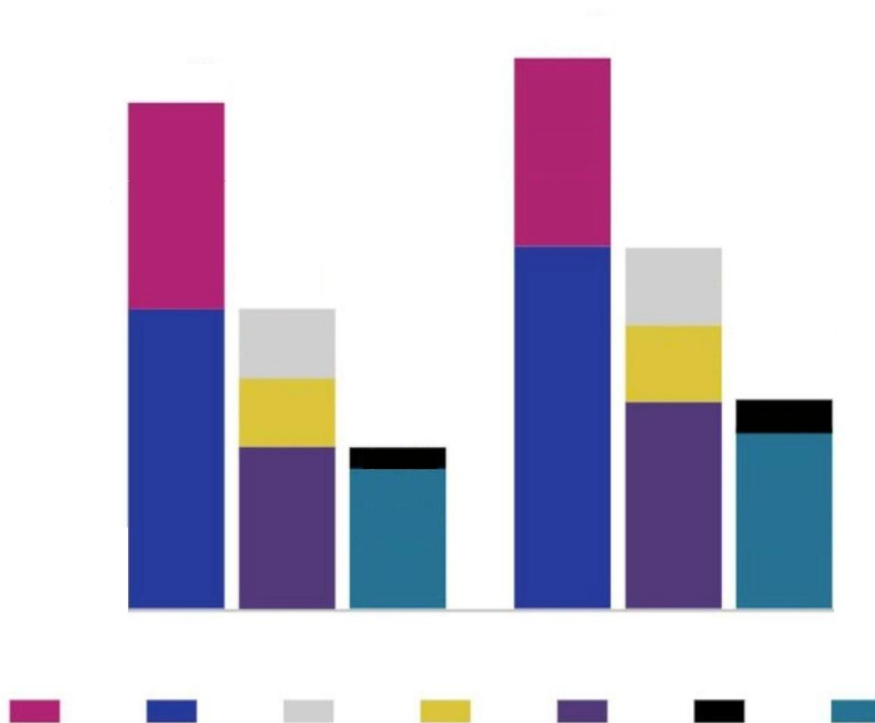


What can you guess? **Structure**



1. **Two groups.** The data has a top-level split into two groups.

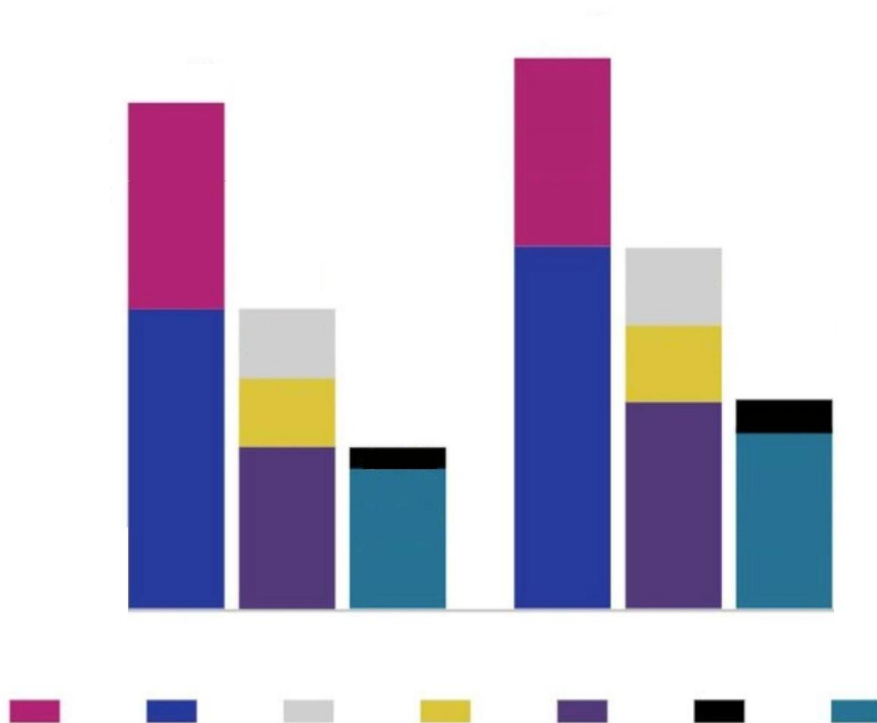
What can you guess? **Structure**



1. **Two groups.** The data has a top-level split into two groups.

2. **Three columns per group.** Within each group, the data is further split into 3 parts, corresponding to the 3 columns.

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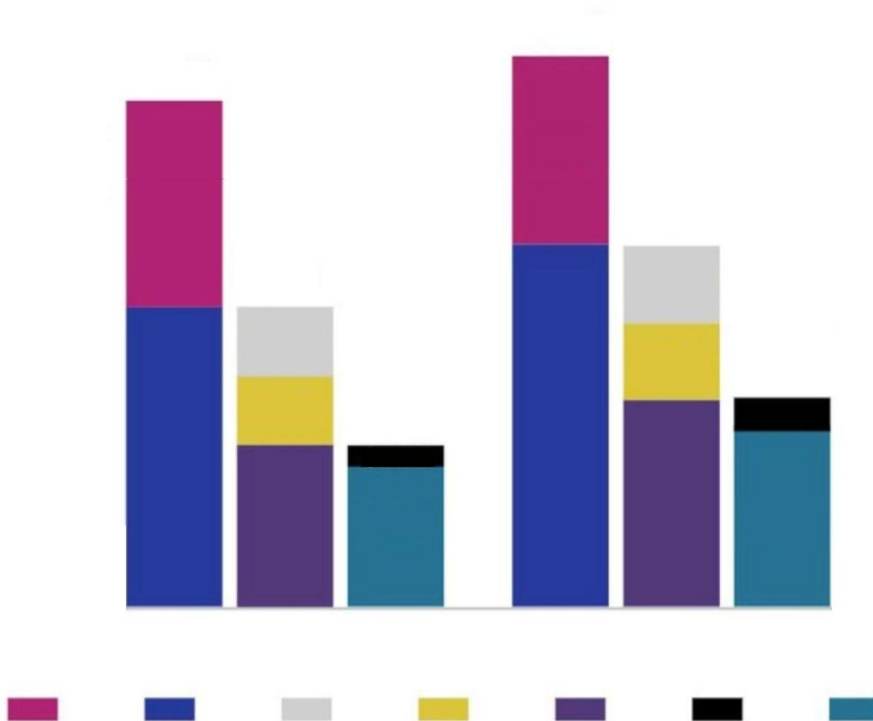


1. **Two groups.** The data has a top-level split into two groups.

2. **Three columns per group.** Within each group, the data is further split into 3 parts, corresponding to the 3 columns.

3. **Variable subparts.** Within each part, there are a variable number of subparts, each given a unique colour.

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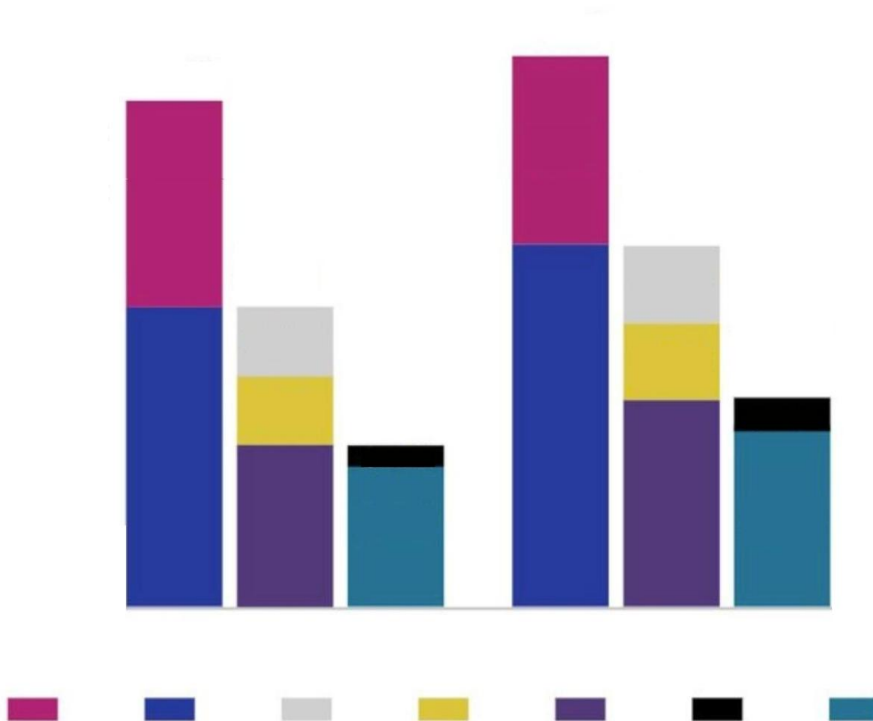
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4. **Seven colours.** The colour legend suggests that each group's data are split into 7 subparts, so I'm guessing that the 7 subparts are aggregated into 3 parts.

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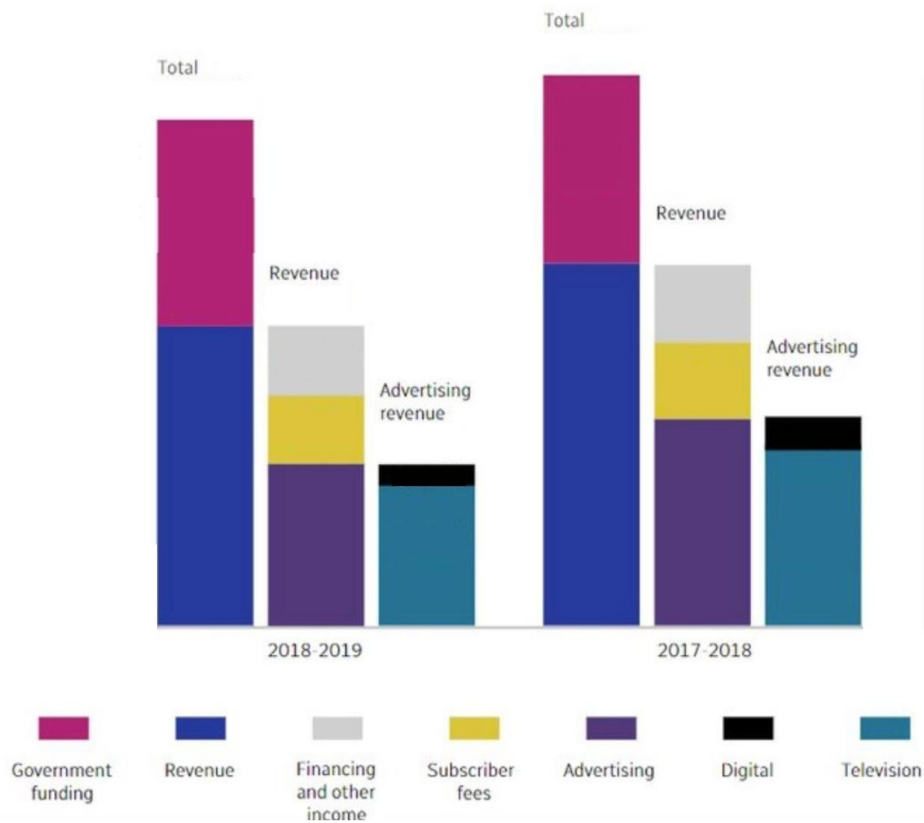
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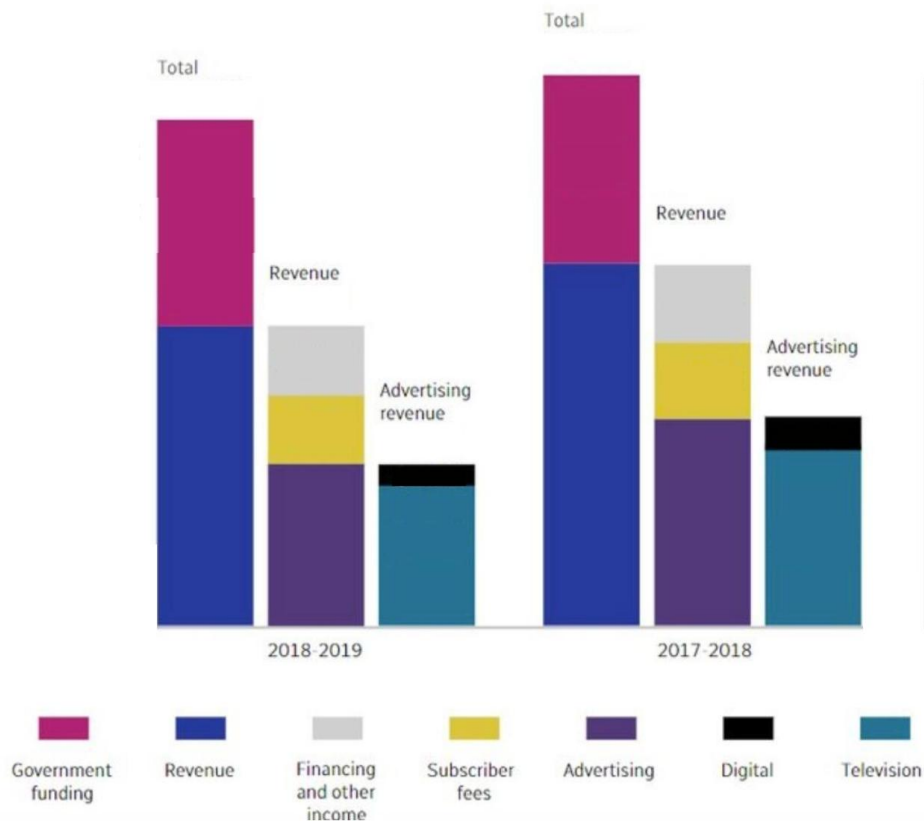
4. **Seven colours.** The colour legend suggests that each group's data are split into 7 subparts, so I'm guessing that the 7 subparts are aggregated into 3 parts.

5. **Stacked column chart.** The core chart form is a stacked column chart with absolute values so relative proportions within each column (part) are important.

What can you guess? **Labels & Comparisons**



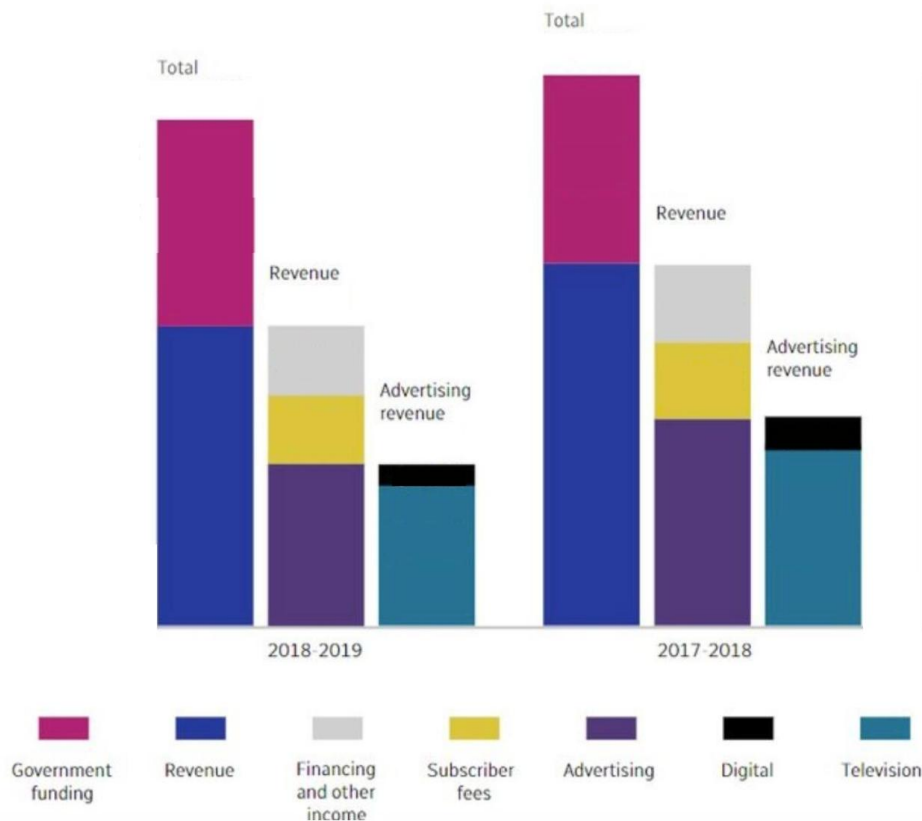
What can you guess? **Labels & Comparisons**



1. No cross-column comparison.

Comparing across columns is not supported because each column has its own total value.

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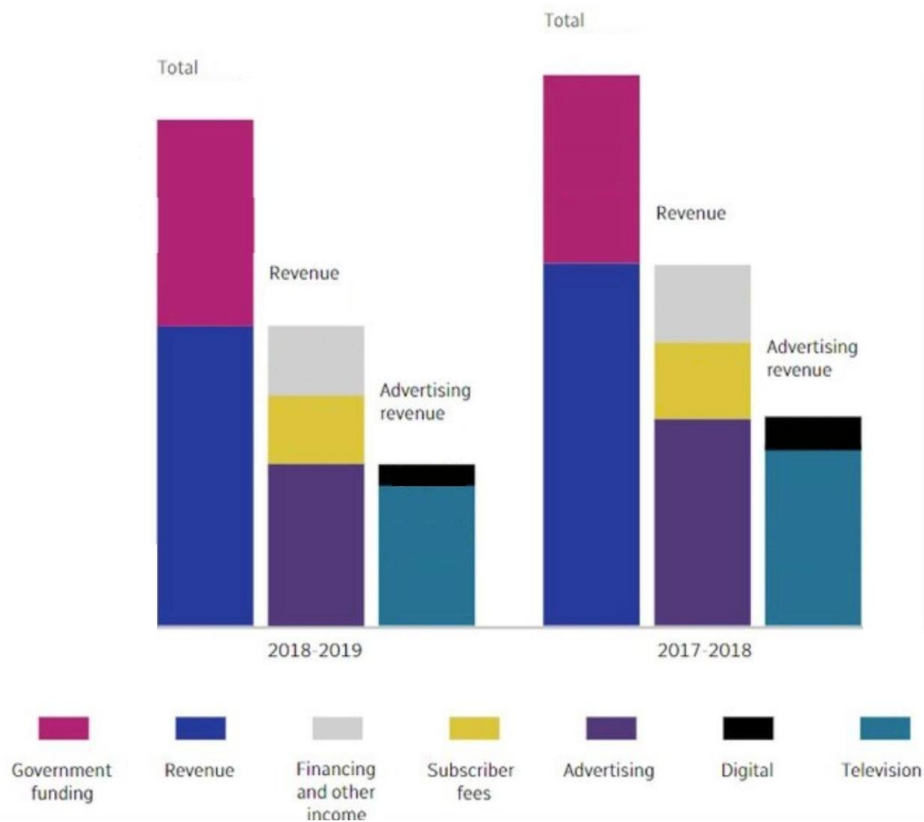
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Comparing same-colour blocks across the two groups is possible. It's easier to compare their absolute values but harder to compare the relative values (proportions of total).

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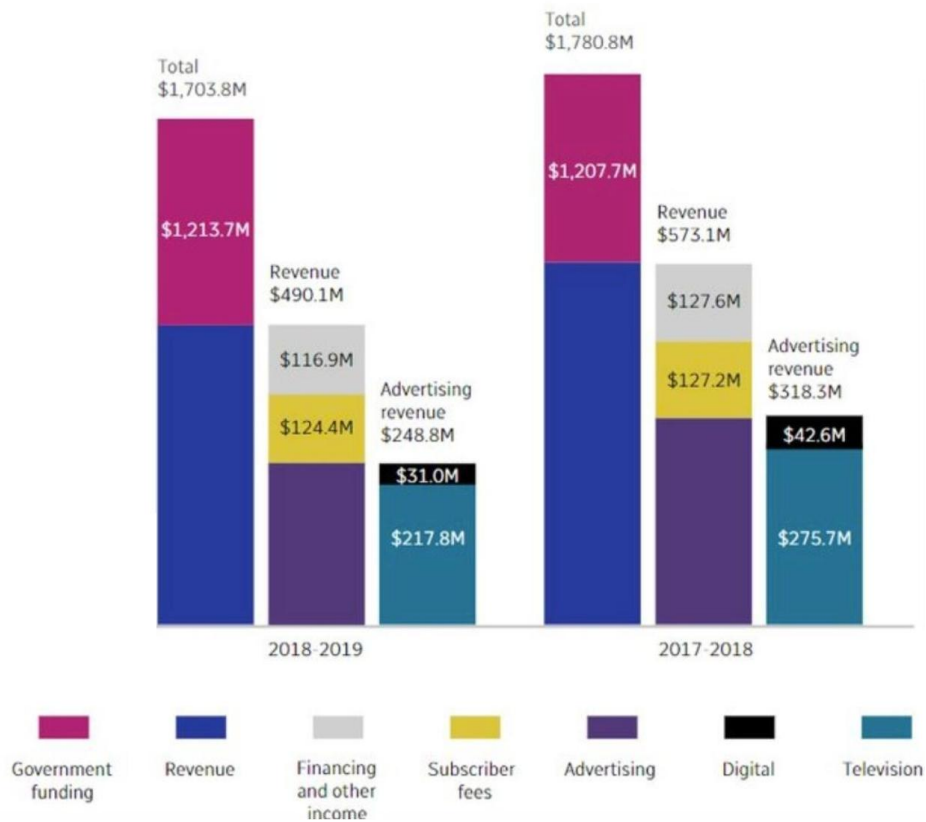
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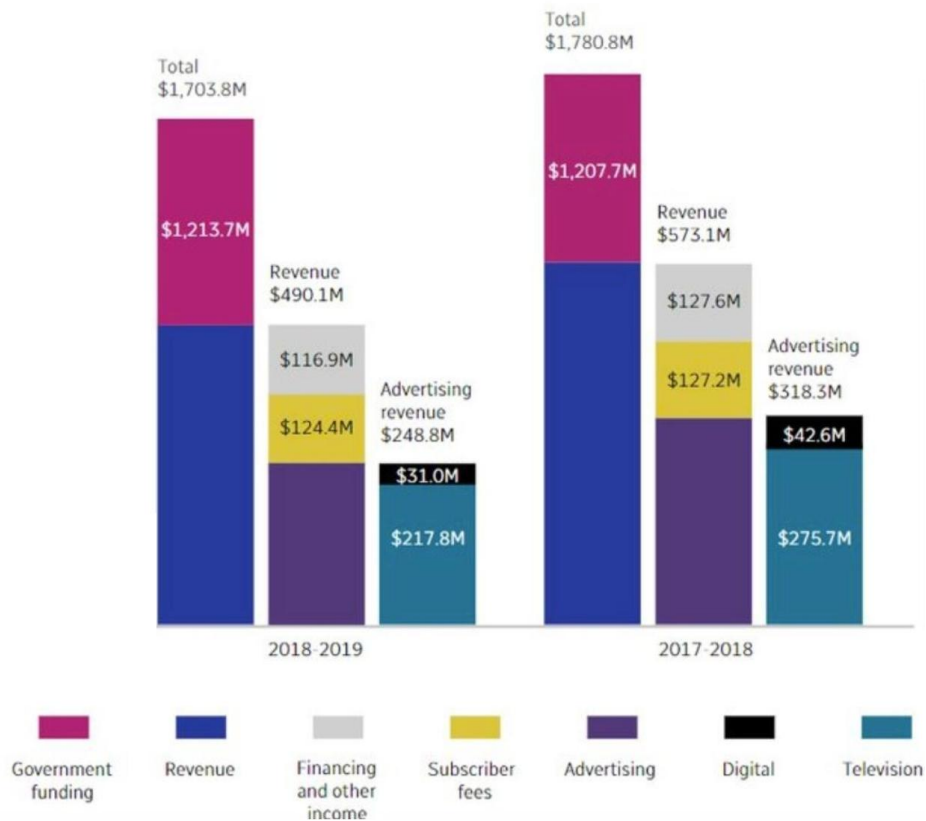
3. Time direction. If the two groups are time periods, I'd guess that the group on the left is the earlier time period, and the one on the right is the later time period.

What can you guess? **Values & Context**



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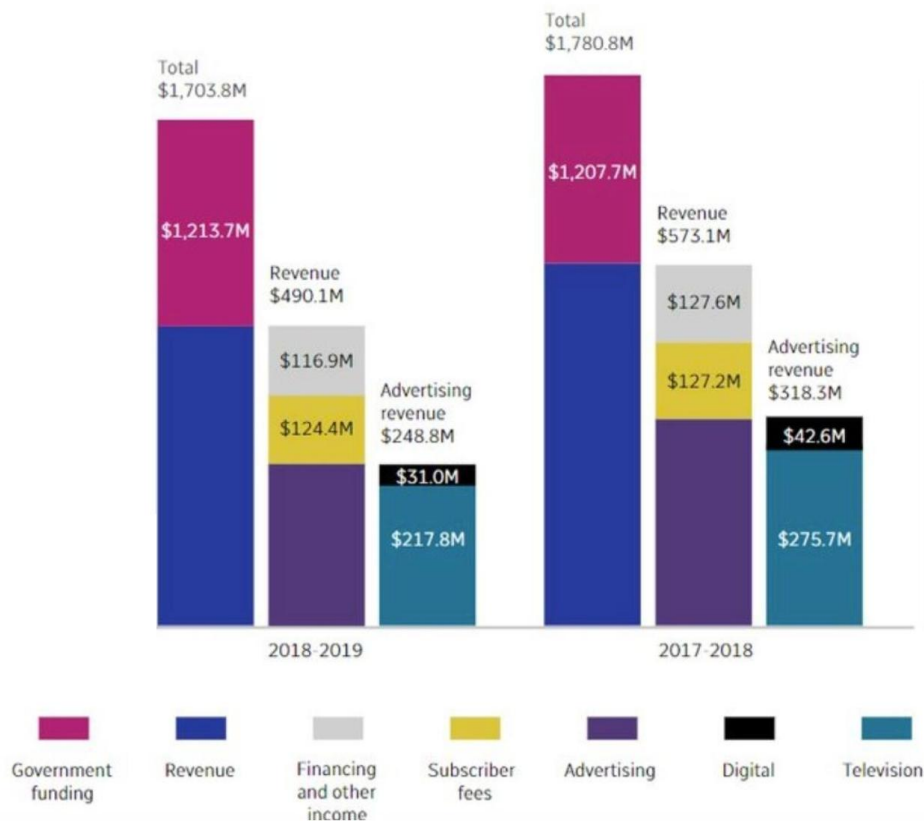
1. **CBC revenues.** Now we see that the chart concerns revenues of a business over two years.



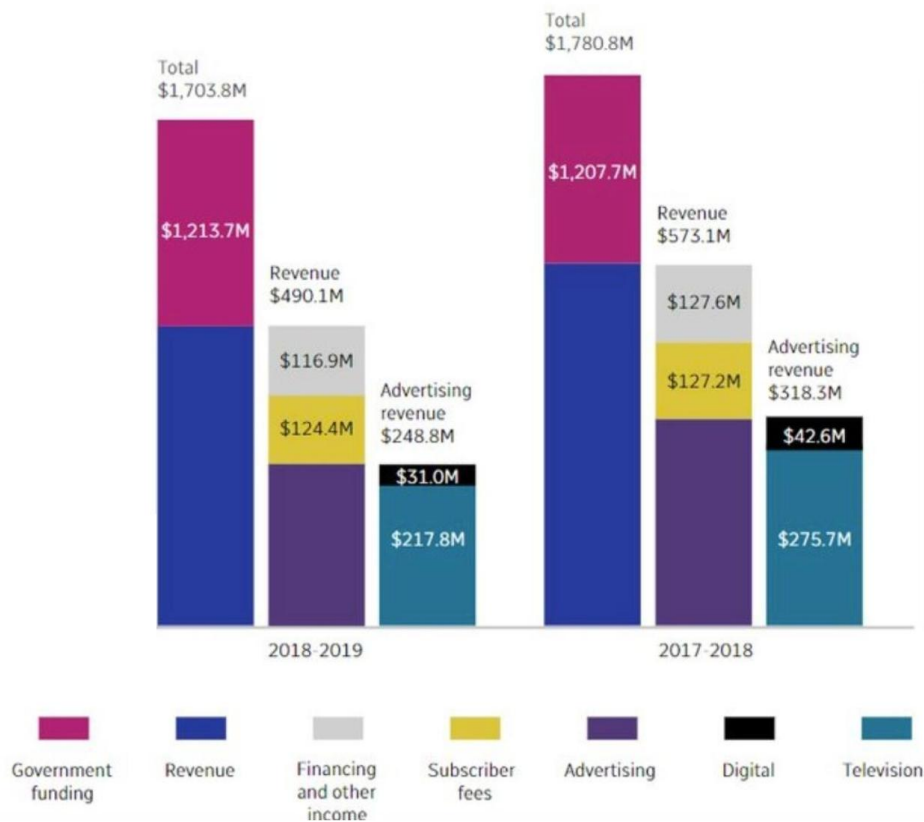
What can you guess? **Values & Context**

1. **CBC revenues.** Now we see that the chart concerns revenues of a business over two years.

2. **Counter-intuitive time order.** My guess on the direction of time was wrong. The more recent year is placed on the left, counter to convention.



What can you guess? **Values & Context**

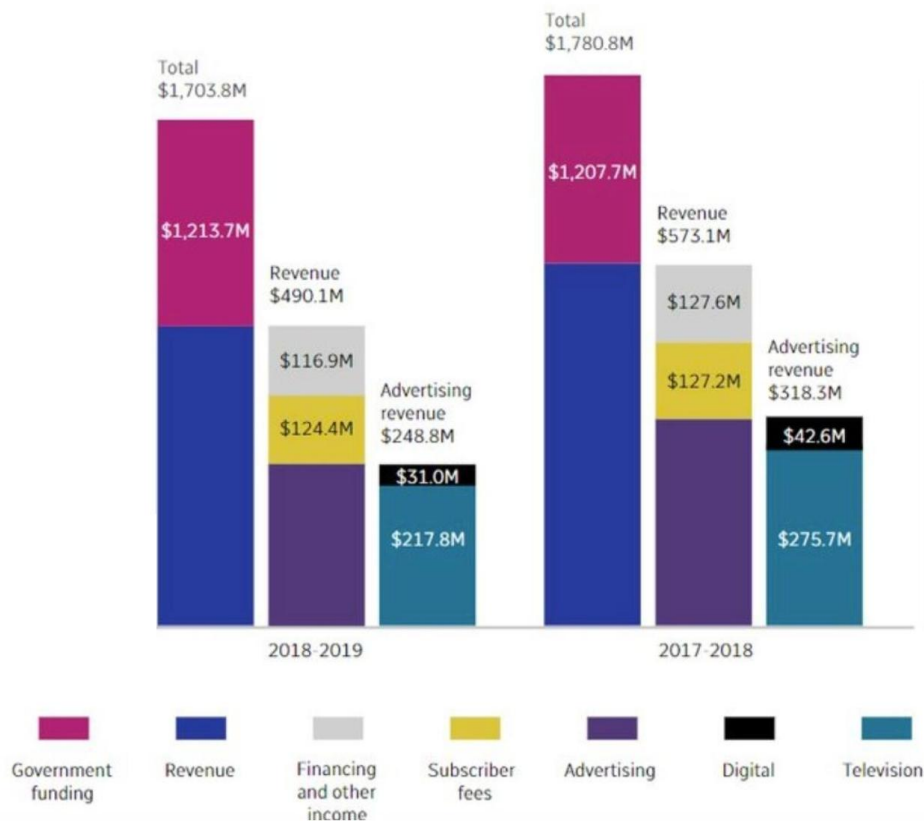


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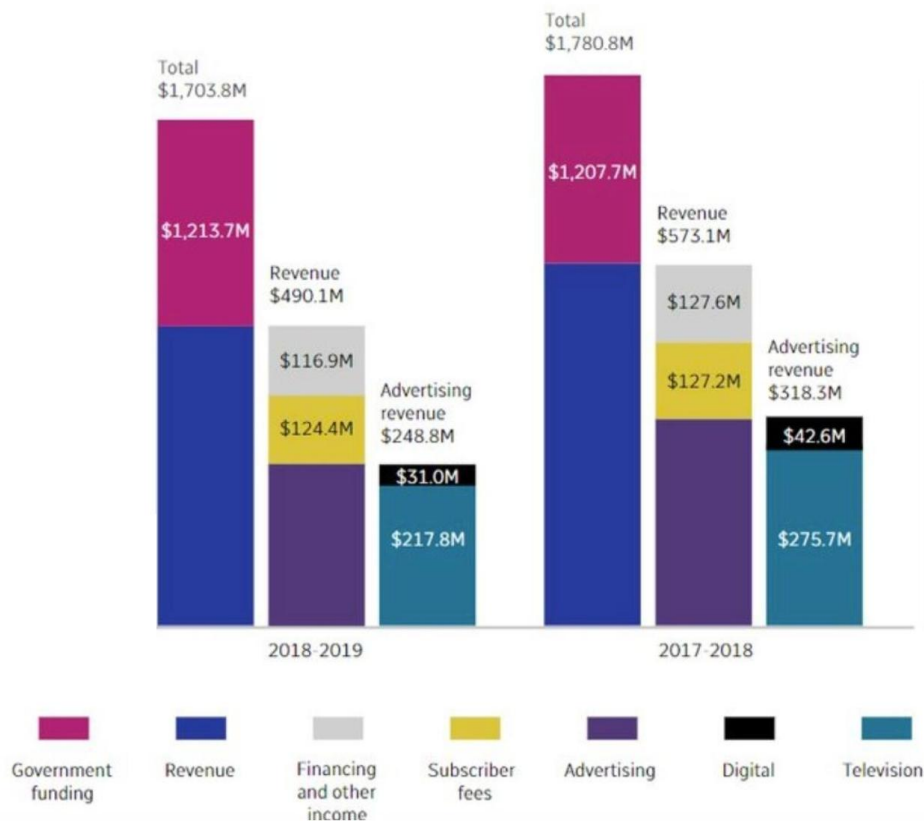
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4. **Overlapping legend categories.** On closer inspection, the 7 colours do not represent 7 independent components of this entity's funding. The categories listed in the colour legend overlap.

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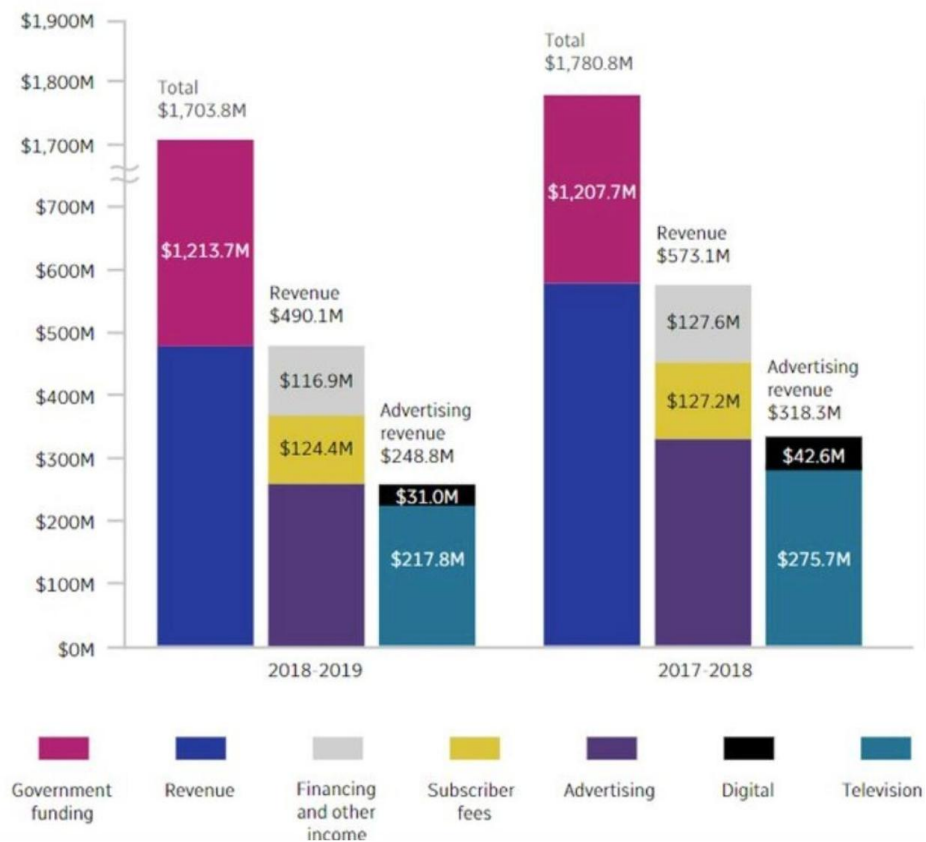
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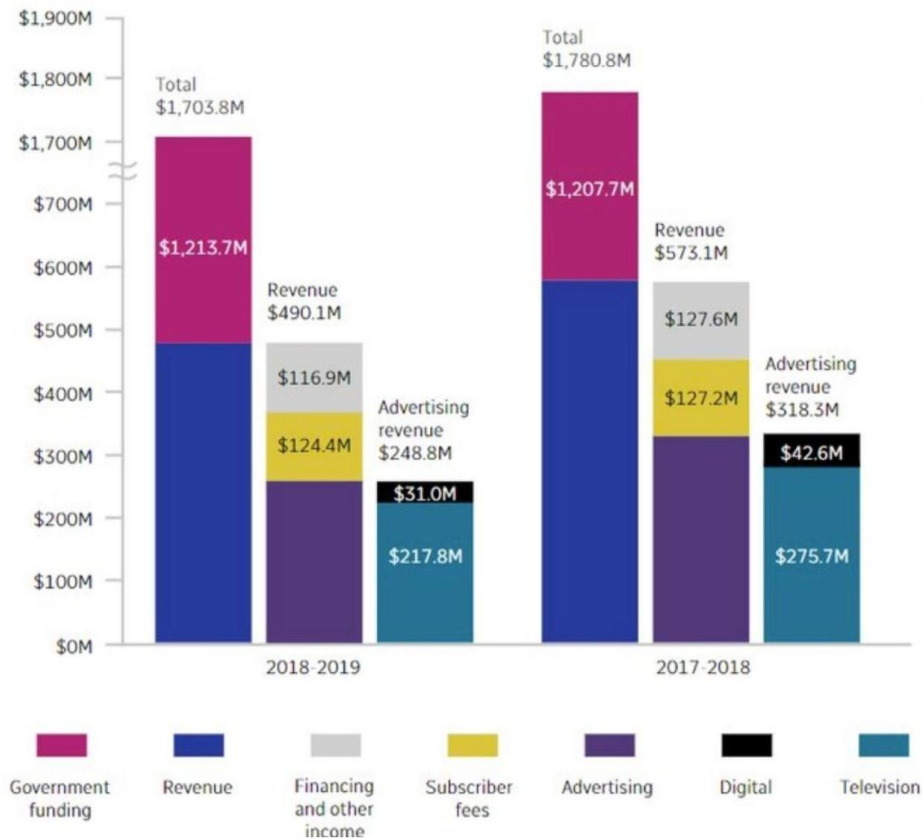
4. **Overlapping legend categories.** On closer inspection, the 7 colours do not represent 7 independent components of this entity's funding. The categories listed in the colour legend overlap.

5. **Nested columns.** The 3 columns within each year-group are nested. Column 2 breaks down revenues into 3 parts; column 3 breaks down advertising revenues into two parts. No visual cues help readers understand how the three columns relate to each other.

What can you guess? **Scale & Hidden Issues**

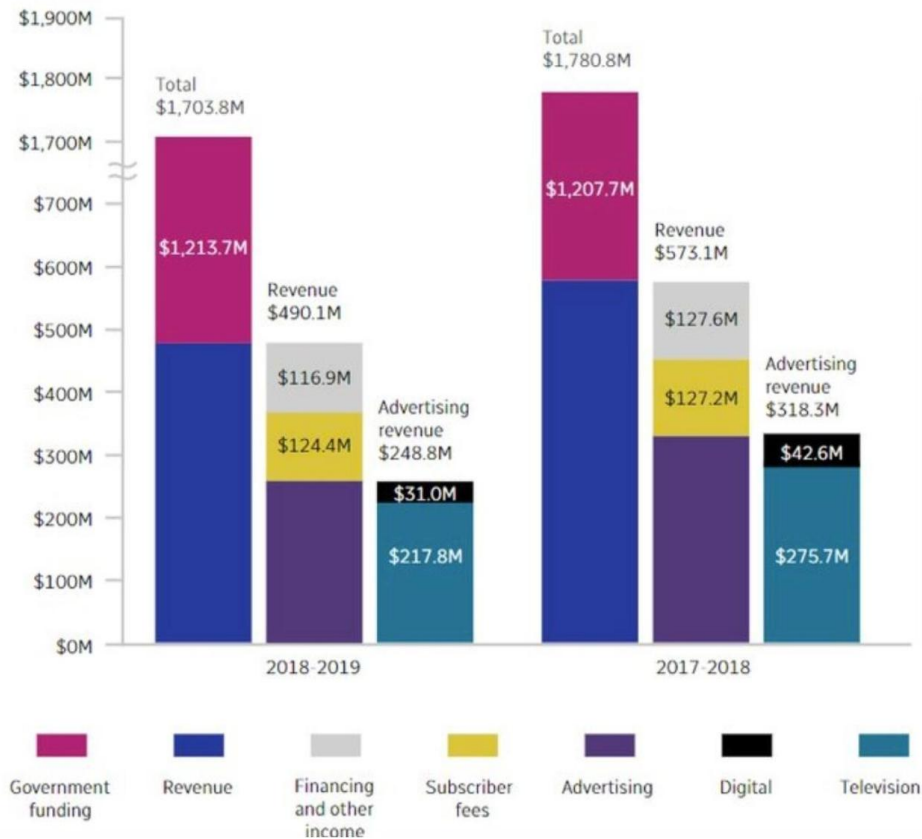


What can you guess? **Scale & Hidden Issues**



1. **Missing labels on largest segments.** Because of the nested structure, two of the column segments (the sums of subparts) are not labelled — meaning the largest parts are the ones without labels while the smaller, less influential subparts are labelled.

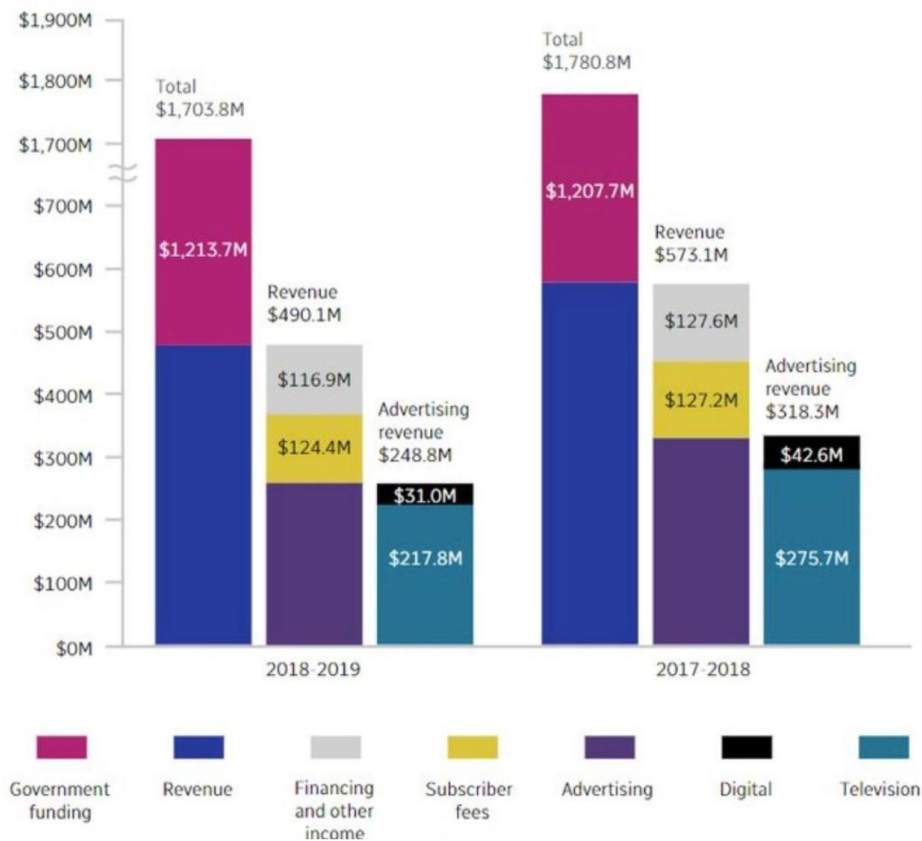
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2. **Labelling rule.** The system of labelling can be described as: label everything that is not further broken down into parts on the chart.

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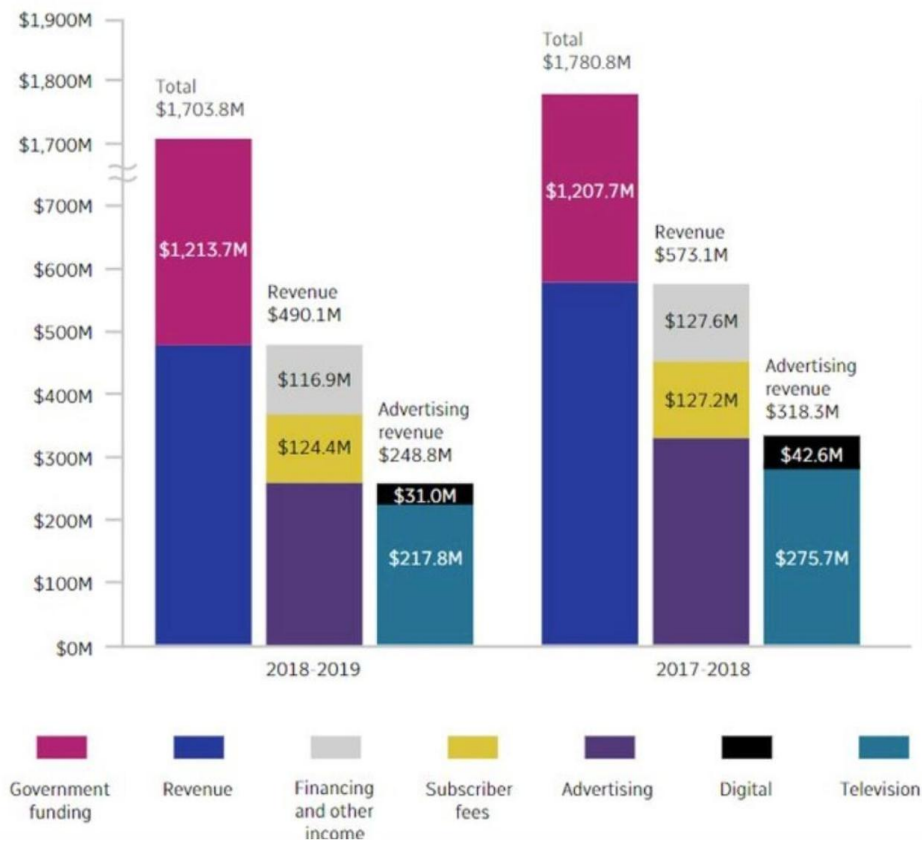


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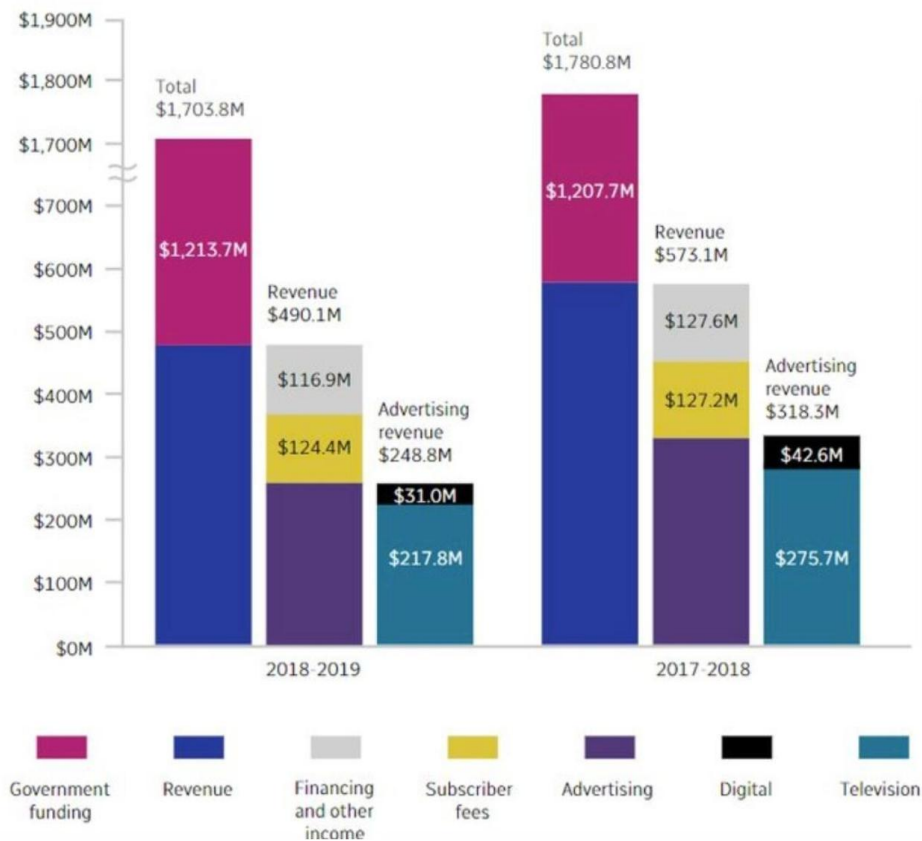
3. **Columns not drawn to scale!** The pink segment is well above \$1 billion yet is roughly the same size as the \$250M third column. These columns are not drawn to scale.

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5. **Key insight hidden.** The focus of the chart is to show what proportion of CBC's funds come from the government. The only way to figure that out is to copy out the data and divide. It's roughly $1.2/1.7 = 70\%$.

Key Takeaways

1. **Structure before data.** Colour, grouping, and chart form tell you a great deal before a single label is revealed.
2. **Labelling choices signal intent.** What is labelled — and what is not — reflects the designer's priorities, often inadvertently.
3. **Nesting without visual cues confuses readers.** When columns are nested but share the same visual treatment, readers cannot reconstruct the hierarchy.
4. **Axis breaks are deceptive.** A discontinuous axis can hide data that is larger than everything else shown — always check the scale.
5. **The chart should answer its own question.** If the point is government funding share (~70%), that number should be visible at a glance, not require a manual calculation.

